

NEUR 411 · NEUROETHOLOGY · FALL 2026

Lecture 11

Insect vision I: the fly optomotor response and the Reichardt elementary motion detector

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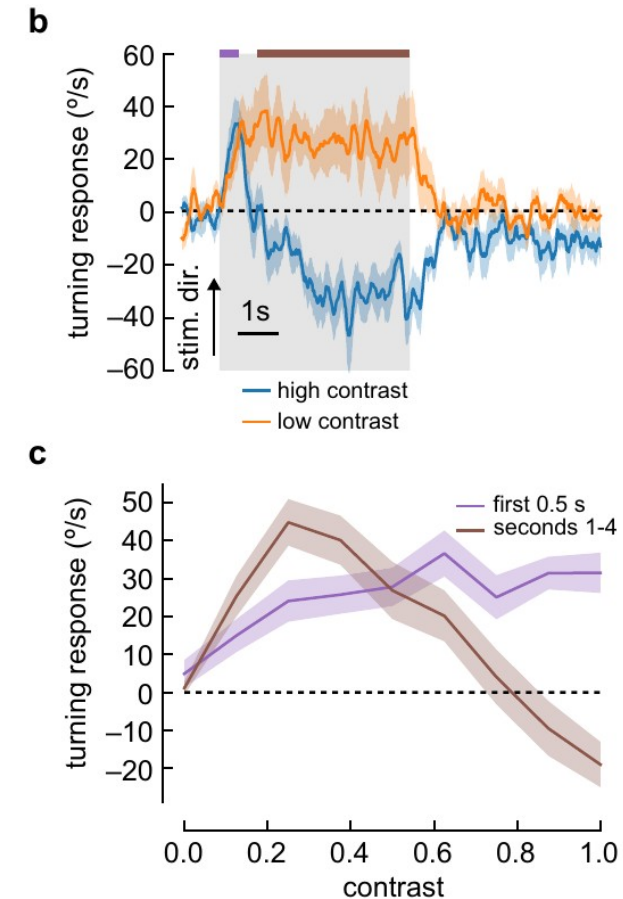
Photo: *Drosophila melanogaster*, the fruit fly used in the motion-vision experiments.

Flies turn with rotating visual patterns

The **optomotor response** is a turn in the direction of a rotating visual pattern. Such turns can reduce **retinal slip**, the movement of the environmental image across the eye during unintended body rotation. Turning direction therefore links visual motion to a measurable motor response.

Mano and colleagues measured tethered *Drosophila melanogaster* walking on an air-supported ball. A panoramic display covered 270°; ball rotation provided turning velocity in degrees per second while the fly remained at a fixed position.

At contrast 0.25, a sinusoidal grating with wavelength 60° moving at 1 Hz elicited sustained turning with the pattern. At contrast 1.0, turning changed direction after approximately 1 s. Optomotor behavior depends on stimulus conditions and elapsed time.



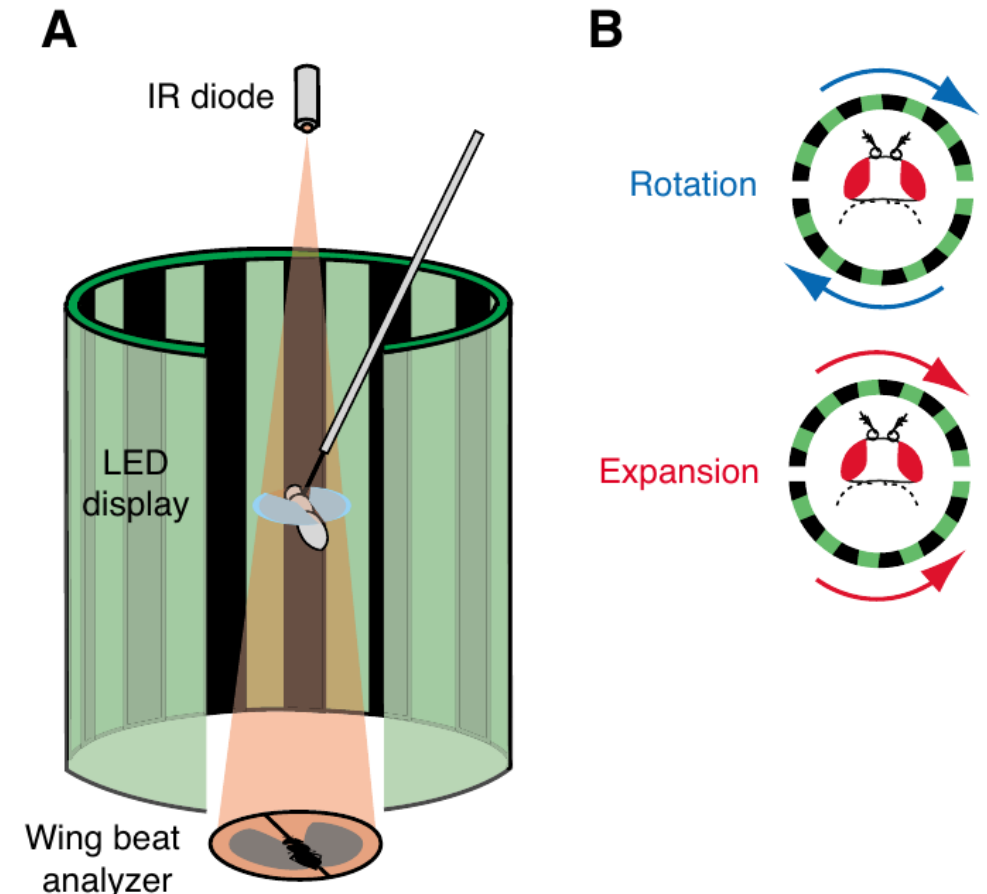
Mano et al. (2023), Fig. 1(b, c). Contrast changes turning during a 5-s rotating pattern.

Controlled optic flow separates rotation from expansion

Optic flow is the pattern of image motion across the visual field. Duistermars and colleagues compared panoramic rotation with an expansion cue in tethered flying *Drosophila*. Rotation carried stripes in one direction around the fly; expansion reversed motion in the rear hemisphere.

Infrared light cast wing shadows onto sensors. The difference between left and right wingbeat amplitudes, ΔWBA , provided a steering measure proportional to yaw torque, the rotational force about the vertical body axis. The recorded voltage was a calibrated proxy for steering, not a body angle.

Each 3-s test followed closed-loop control of a vertical stripe. The expansion stimulus was an approximation to translation, with an abrupt velocity reversal between hemispheres, rather than the smooth flow field of natural forward flight.



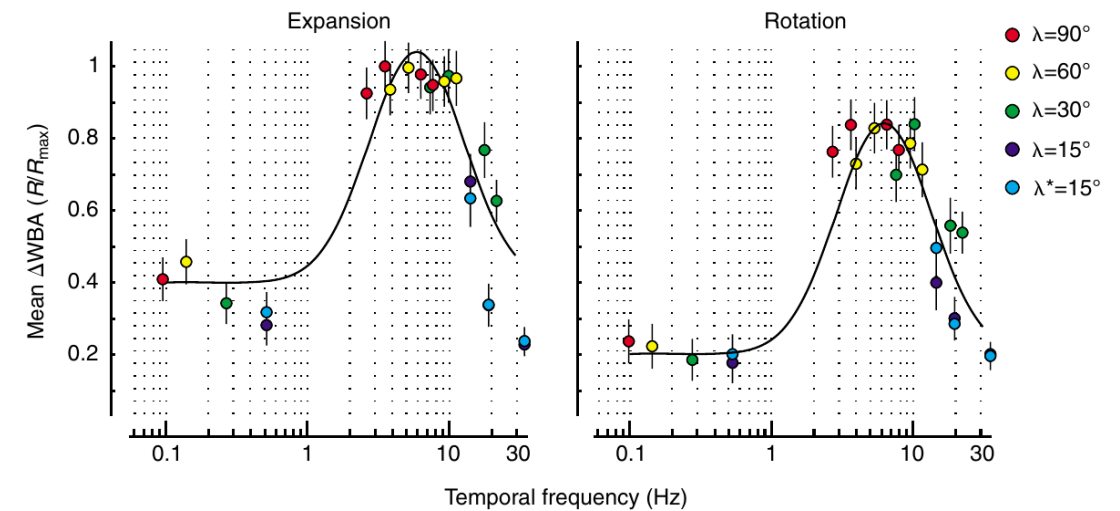
Duistermars et al. (2007), Fig. 1. Tethered-flight apparatus and two panoramic flow fields.

Motion tuning depends on space and time

A grating's **spatial wavelength** is the angular distance between repeating stripes. Its **temporal frequency** is the number of pattern cycles passing a retinal position each second. A pattern's speed alone therefore does not specify the stimulation reaching the visual system.

Duistermars and colleagues varied wavelengths from 15° to 90° and pattern speeds from 10 to $431^\circ/\text{s}$. These combinations sampled approximately 0.11–35.4 Hz while the same flies encountered both expansion and rotation.

Expansion generally elicited stronger steering than rotation, but both responses plateaued at approximately 3–10 Hz and declined near 30 Hz. Comparable temporal tuning is consistent with shared elementary motion processing; different response magnitudes do not establish separate local detectors.



Duistermars et al. (2007), Fig. 3. Expansion and rotation share similar temporal tuning.

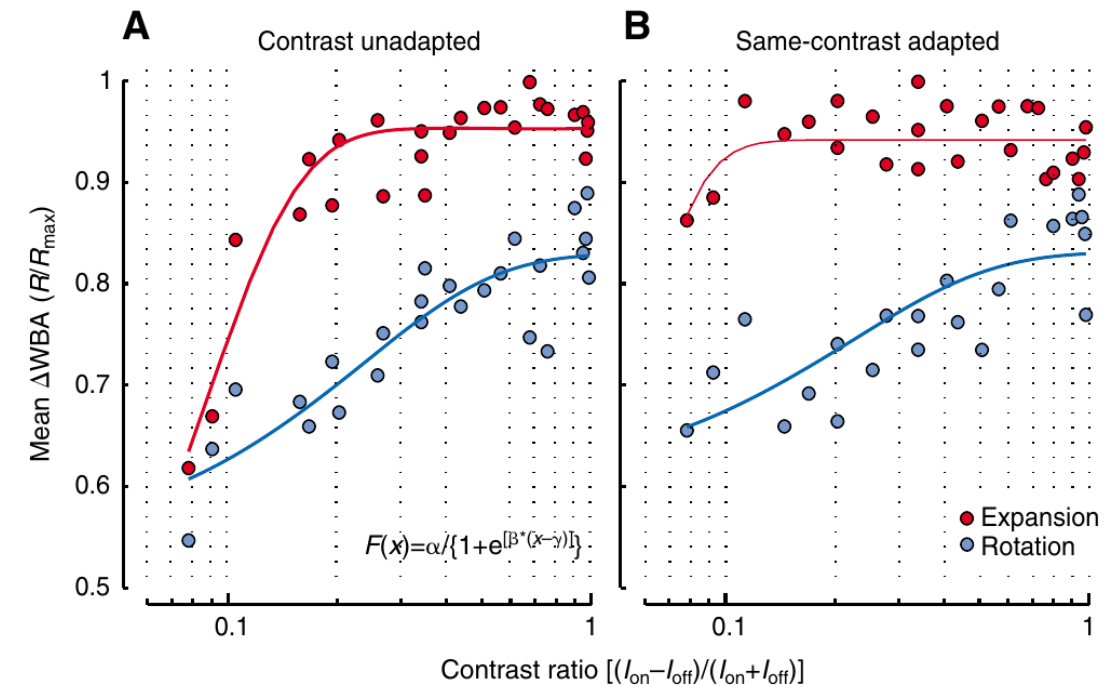
Contrast changes the strength of optomotor turning

Contrast quantifies the difference between bright and dark pattern regions relative to their combined intensity.

Duistermars and colleagues varied contrast while holding wavelength at 30° and speed at $232^\circ/\text{s}$, separating pattern visibility from changes in its motion.

Before an unadapted test, the fly controlled a single stripe. Before an adapted test, it experienced the wide-field test contrast for 7 s. Thus the comparison changed recent visual exposure without changing the subsequent moving pattern.

Expansion steering saturated at lower contrasts than rotational steering in the unadapted condition. Prior contrast exposure altered sensitivity and made the response relationships more nearly linear. Behavioral gain, the response change per stimulus change, depends on recent stimulation as well as motion direction.



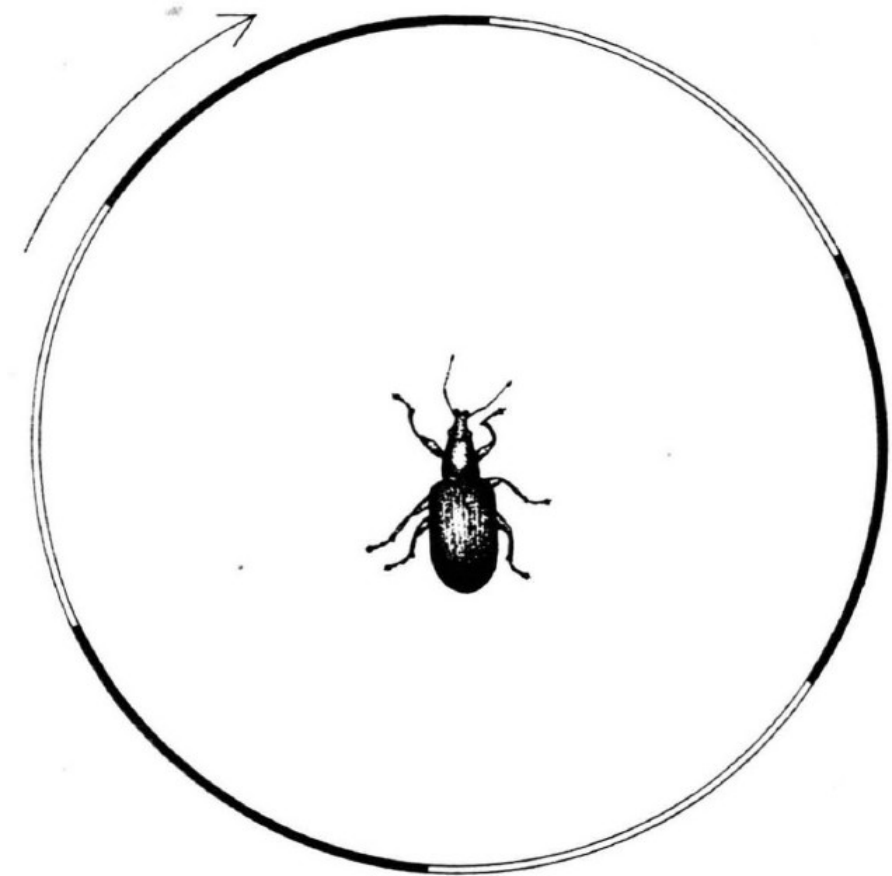
Duistermars et al. (2007), Fig. 7. Prior contrast exposure changes steering sensitivity.

Reichardt's original experiments used a beetle

Hassenstein and Reichardt developed their motion-detection account from the weevil *Chlorophanus*, not from *Drosophila*. A striped rotating surround elicited a compensatory turning response, linking the sequence of visual stimulation to a directional behavioral output.

Their 1956 analysis separated the timing, order and sign of local brightness changes. A motion signal required a relationship between neighboring visual inputs: a single location changing brightness did not uniquely specify which direction an object traveled.

The **elementary motion detector** is a local computation comparing signals from different visual positions over time. Its historical derivation described operations sufficient to explain behavior. Later fly experiments tested that computation and investigated its neural implementation; the beetle experiments did not identify T4 or T5 neurons.



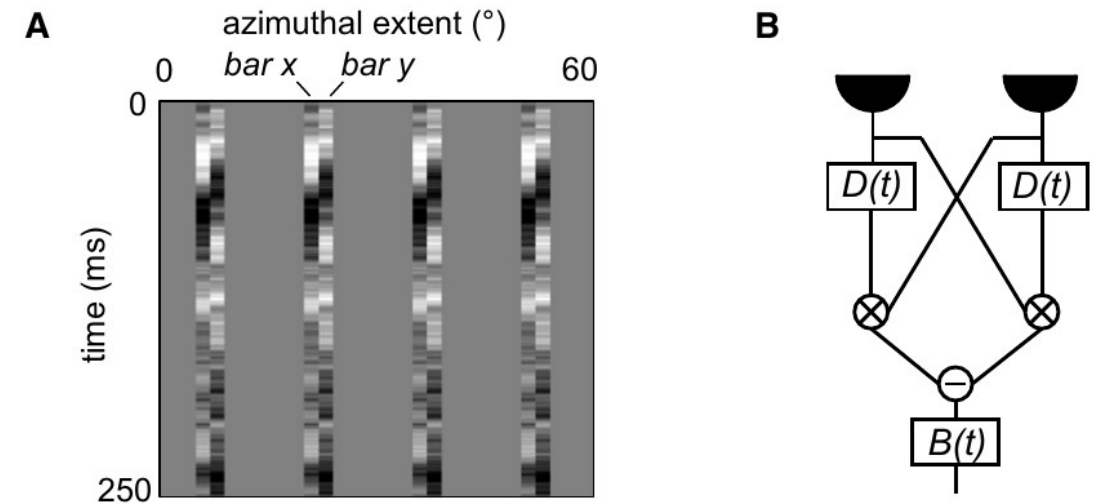
Hassenstein & Reichardt (1956), Fig. 2. A beetle inside the original rotating striped drum.

Sequential stimulation supplies directional information

Apparent motion can arise when adjacent visual positions are stimulated successively, even without a continuously moving object. Reversing the order of stimulation changes the available direction cue while leaving the individual locations unchanged.

Clark and colleagues presented independently varying pairs of 2°-wide bars to flies walking on a ball. The repeated bar pairs supplied local intensity sequences without requiring the experimenter to assign a coherent moving feature to every frame.

Turning was related to correlations between neighboring bar intensities separated by approximately 20–30 ms. A **correlation** here means a joint dependence on the two signals. The timing relationship supports a delayed interaction, rather than a detector responding only to the intensity at one location.



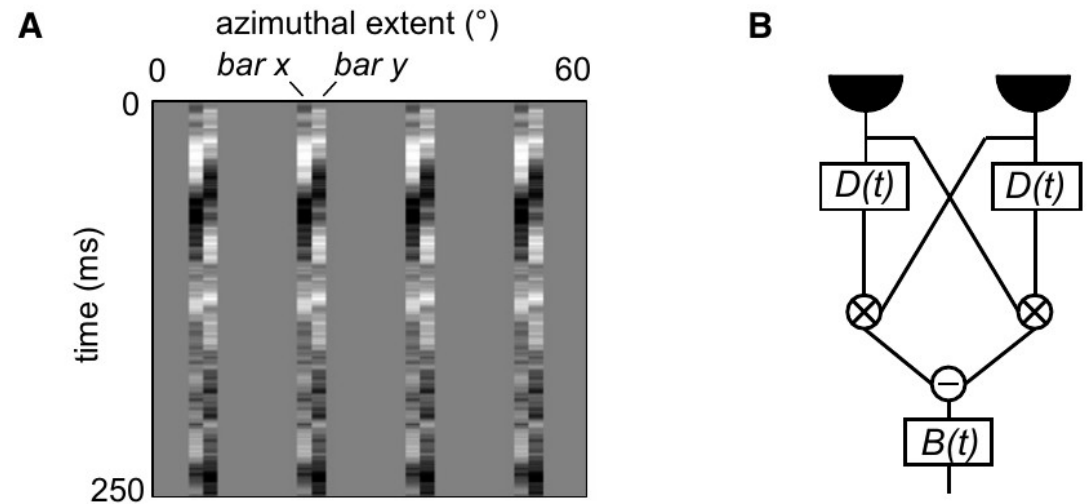
Clark et al. (2011), Fig. 2(A, B). Independent neighboring bars probe a delayed interaction.

A delayed interaction compares neighboring signals

The **Hassenstein–Reichardt correlator** compares two neighboring inputs after applying different temporal filtering to them. One signal is delayed relative to the other, allowing a moving pattern's sequential activation to overlap within the computation.

The delayed and less-delayed signals interact multiplicatively: their joint contribution depends on both inputs, rather than their simple sum. A mirror-symmetric arm reverses which spatial input is delayed, and the two arm outputs are subtracted.

Clark's random-bar experiments supported this correlation structure in fly turning, with a fitted delay filter peaking near 25 ms. The diagram is a computational model. A delay box and multiplier do not, by themselves, identify particular neurons, channels or synapses that implement those operations.



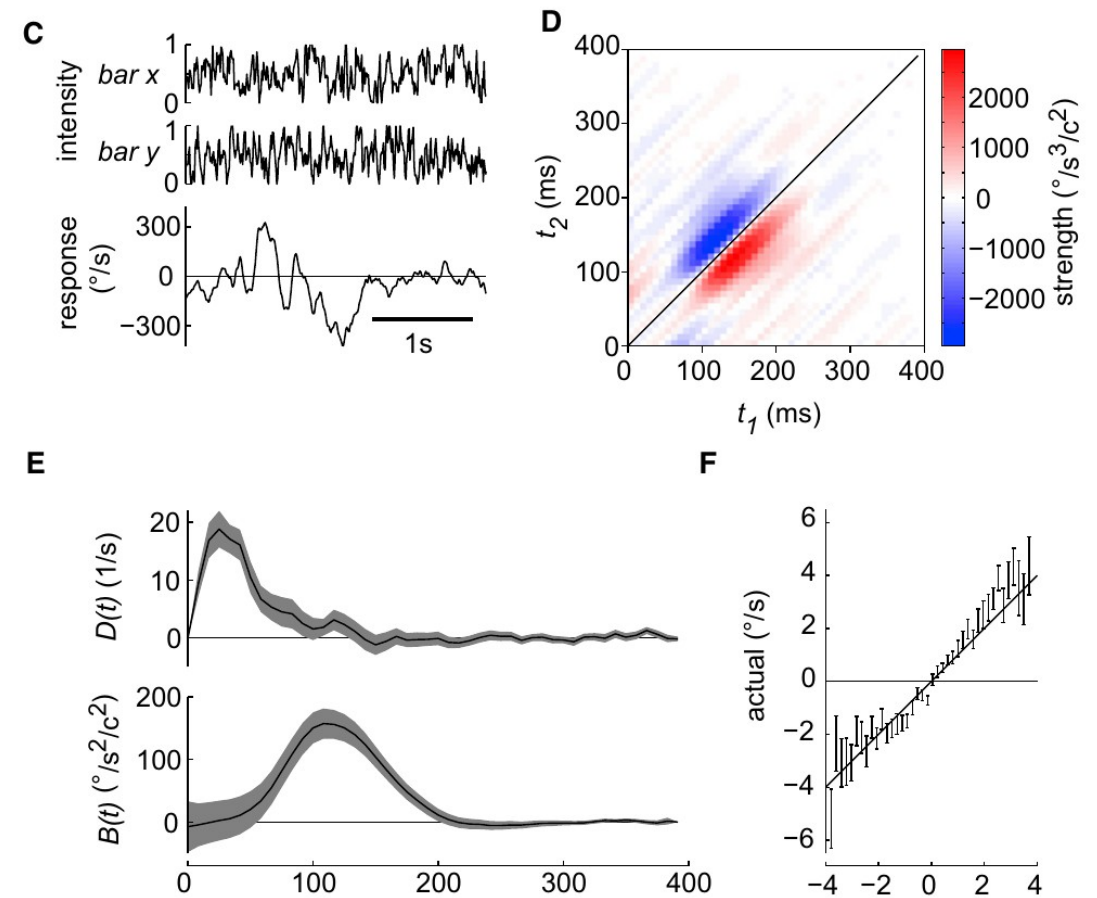
Clark et al. (2011), Fig. 2(A, B). The published correlation detector and its visual inputs.

Opponent detector arms distinguish motion direction

An **opponent** computation subtracts signals favoring opposite alternatives. In the Reichardt detector, the two arms exchange the positions receiving the relative delay. Direction is encoded by their imbalance, rather than by the activity of either arm alone.

Clark and colleagues fitted a two-dimensional temporal kernel, a weighting of earlier input combinations that predicted turning. Neighboring intensities contributed most strongly when separated by 20–30 ms, and the relationship between predicted and measured mean turning was approximately linear.

The fit accounted for only a small fraction of the variability under this weak random-bar stimulus. Agreement with the average directional interaction supports the model's computational structure while leaving substantial behavioral fluctuations and downstream response dynamics unexplained.



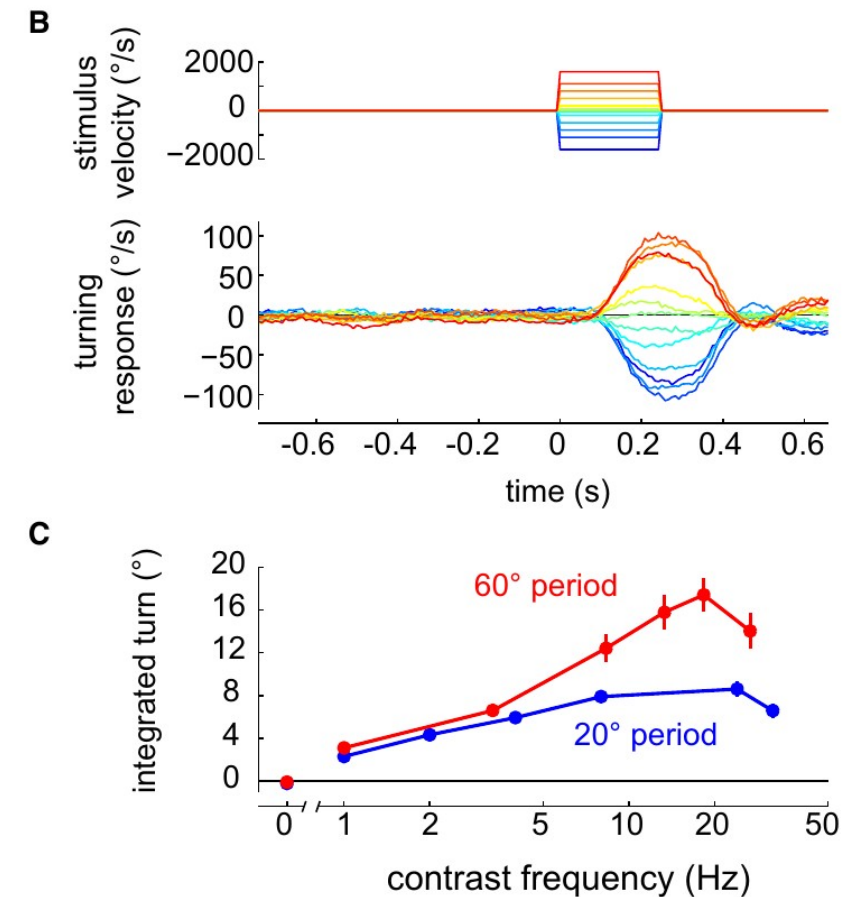
Clark et al. (2011), Fig. 2(C, D, E, F). Measured temporal kernel and predicted versus actual turning.

Correlation detectors do not measure velocity alone

An **angular velocity** reports how many degrees a pattern moves per second. A correlation detector's response also depends on spatial structure and temporal filtering, so it is not a pattern-independent speedometer.

Clark and colleagues measured turning responses to gratings with 20° and 60° spatial periods across multiple temporal frequencies. Turning increased and then declined as temporal frequency rose. Duistermars obtained a comparable band of effective temporal frequencies during tethered flight.

Changing wavelength changes the time between neighboring contrast events at a fixed speed. A detector can therefore respond differently to two patterns moving at the same velocity. Behavioral tuning constrains the overall computation, but it also includes spatial pooling, adaptation and the dynamics of the motor response.



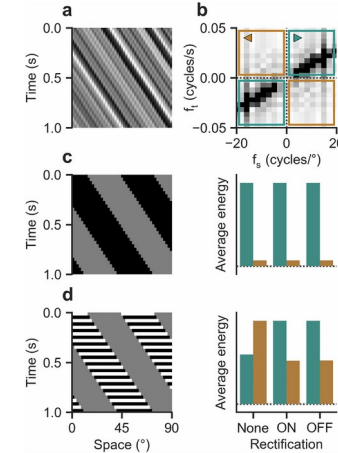
Clark et al. (2011), Fig. 1(B, C). Grating spacing changes the relationship between speed and turning.

Contrast reversals can reverse perceived motion

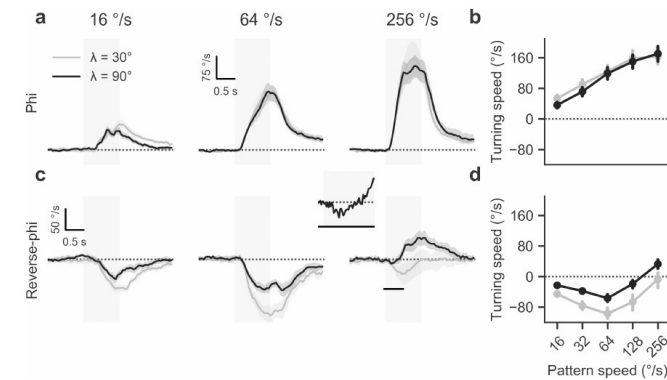
In **phi motion**, a pattern shifts between frames while preserving its contrast. In **reverse-phi motion**, each shift also reverses bright and dark regions. Physical displacement and the direction signaled by the visual system can then disagree.

Leonhardt and colleagues measured tethered walking during discrete pattern shifts. Ordinary phi elicited turning with the pattern. Reverse-phi elicited turning against it at lower velocities, while sufficiently rapid stimulation could reverse the behavioral response again.

Contrast inversion changes the sign of interactions between neighboring signals in a correlation computation. Reverse-phi therefore probes the detector's operations, rather than simply testing whether the fly can track an identifiable object. The illusion's dependence on velocity limits a single universal prediction of opposite-direction turning.



Leonhardt et al. (2017), Fig. 2(a-f). Contrast-preserving and contrast-reversing apparent-motion stimuli.



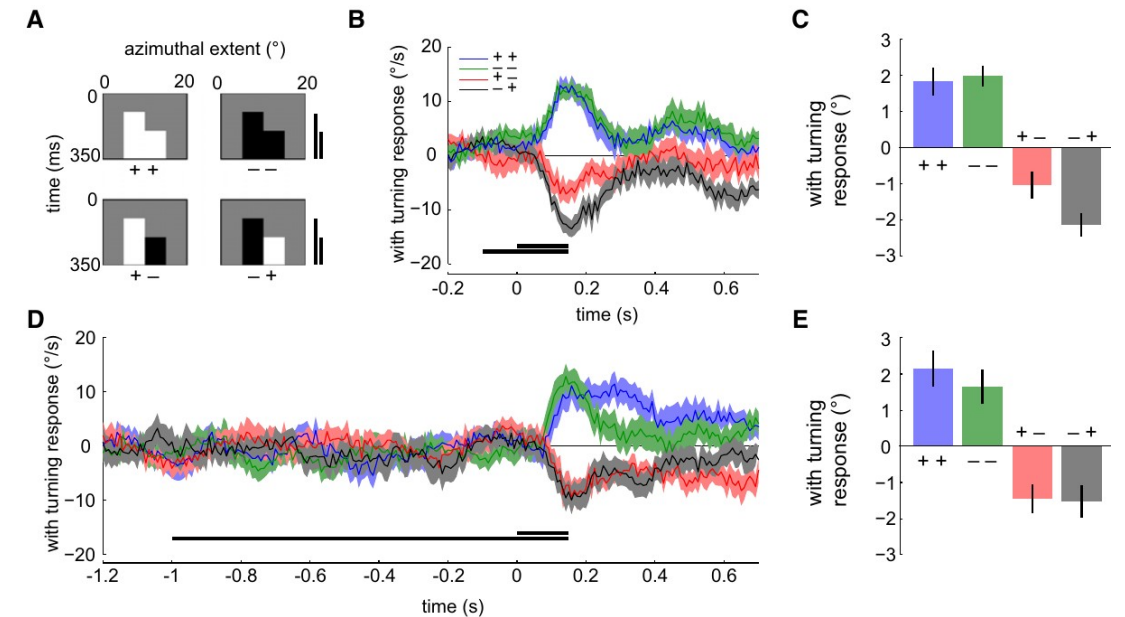
Leonhardt et al. (2017), Fig. 3. Measured turning reverses under reverse-phi stimulation.

Contrast-polarity tests constrain detector architecture

A **contrast polarity** specifies whether local brightness increases or decreases relative to a reference. Two successive neighboring events can therefore form four pairings: bright–bright, dark–dark, bright–dark and dark–bright.

Clark and colleagues presented minimal sequential bar pairs while recording ball-based turning. Flies responded to all four pairings. Contrast-preserving sequences favored one directional sign; contrast-reversing sequences favored the opposite sign under the tested timing conditions.

These results constrain how ON and OFF pathways can implement the correlation computation. Processing bright and dark events separately cannot mean that all opposite-polarity influences are absent. Behavioral sensitivity identifies effective operations, but it does not establish the membrane mechanism or exact synaptic site of each interaction.



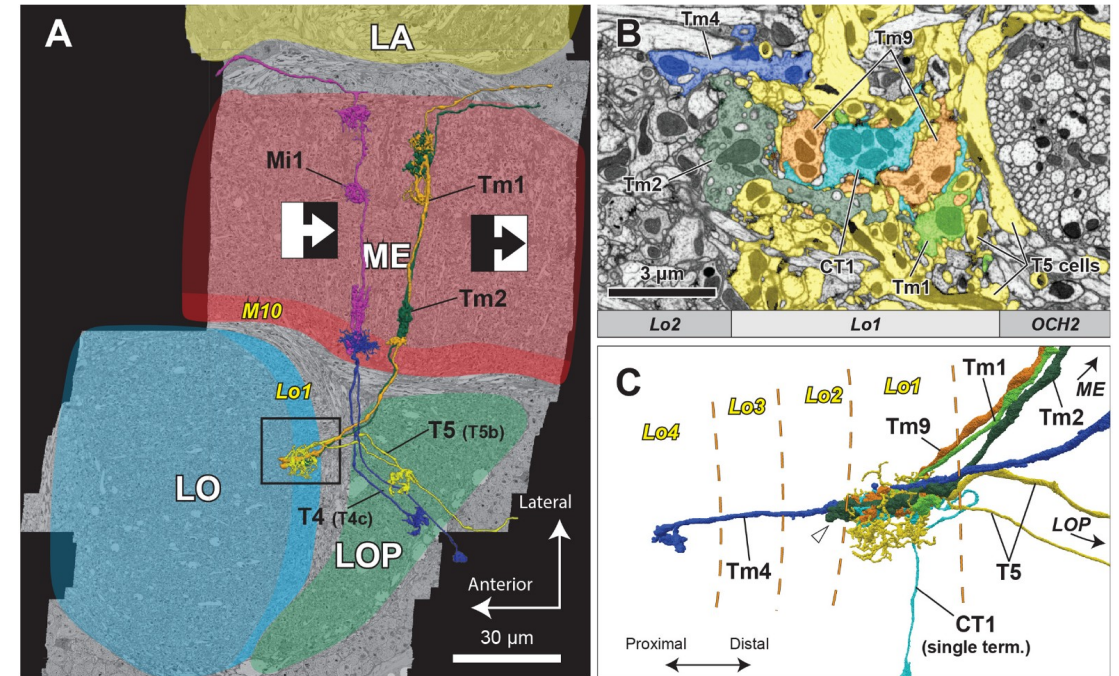
Clark et al. (2011), Fig. 6. All four contrast-pair sequences influence fly turning.

The fly optic lobe preserves visual position

The **optic lobe** contains successive visual processing regions beside the central brain. A **neuropil** is a region rich in neuronal processes and synapses. The fly's major optic neuropils are the lamina, medulla, lobula and lobula plate.

Retinotopy preserves the neighborhood relationships of visual positions within neural tissue. Repeated **columns** carry signals associated with neighboring locations, so local motion can be computed by comparing inputs from adjacent columns rather than losing spatial identity immediately.

Shinomiya and colleagues reconstructed neurons and synapses with focused-ion-beam scanning electron microscopy. T4 dendrites occupy medulla layer M10, whereas T5 dendrites occupy lobula layer Lo1. Their axons converge in the lobula plate. Anatomical position distinguishes the two processing routes but does not establish their temporal response properties.



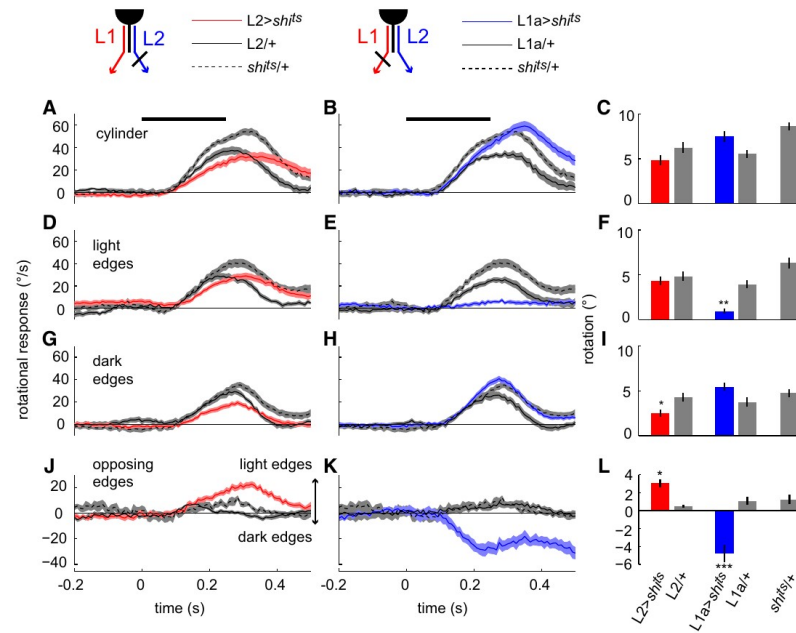
Shinomiya et al. (2019), Fig. 1. Reconstructed neurons traverse the optic-lobe neuropils.

Lamina pathways divide luminance-change signals

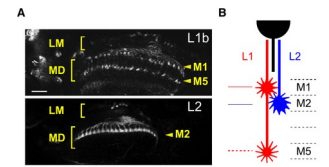
Luminance is the light intensity represented at a visual location. Lamina neurons L1 and L2 relay signals from the outer photoreceptors R1–R6 into the medulla. Their axon terminals occupy different medulla layers. Photoreceptor **histamine**, a neurotransmitter, activates chloride channels in L1/L2 and inverts the light response: increased brightness reduces terminal calcium.

Clark and colleagues used **shibire**, a temperature-sensitive interference with synaptic vesicle recycling, to suppress output from selected neurons. They compared experimental flies with driver-only and effector-only controls while presenting isolated light edges, dark edges or opposing edges.

L1-associated processing favored light-edge motion, whereas L2-associated processing favored dark-edge motion. Both pathways retained overlapping computations, and their terminal calcium signals were not simply opposite-polarity channels. The behavioral division cannot be inferred solely from the sign of a lamina neuron's response to light.



Clark et al. (2011), Fig. 3. Silencing L1 or L2 changes edge-polarity steering.



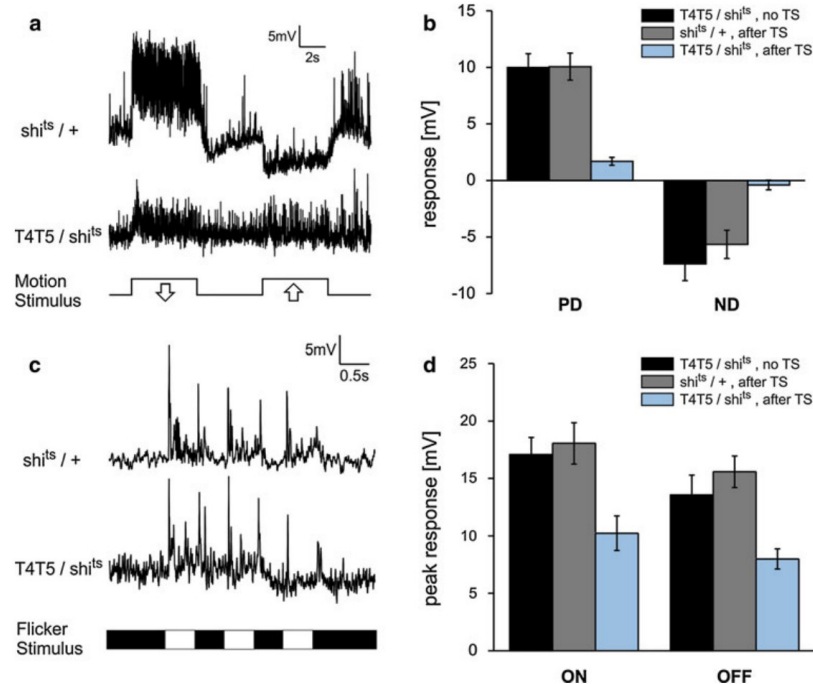
Clark et al. (2011), Fig. 4(A, B). L1/L2 terminals.

T4 and T5 provide local direction-selective signals

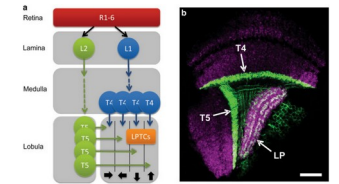
T4 and **T5** are repeated small-field neurons linking earlier visual pathways to the lobula plate. A **direction-selective** neuron responds differently to opposite motion directions through the same region of visual space. T4 and T5 supply direction-selective signals to larger visual interneurons.

Schnell and colleagues recorded membrane voltage from lobula plate tangential cells while suppressing T4/T5 output with temperature-sensitive shibire. A tangential cell extends processes across many visual columns, allowing comparison between a local input population and a wide-field response.

Motion-evoked depolarization and hyperpolarization were largely abolished during the block, whereas flicker responses remained. A **flicker** changes brightness without ordered spatial displacement. The selective loss supports a necessary motion input from T4/T5 rather than complete failure of the recorded neuron.



Schnell et al. (2012), Fig. 2(a-d). Motion responses disappear while flicker responses persist.



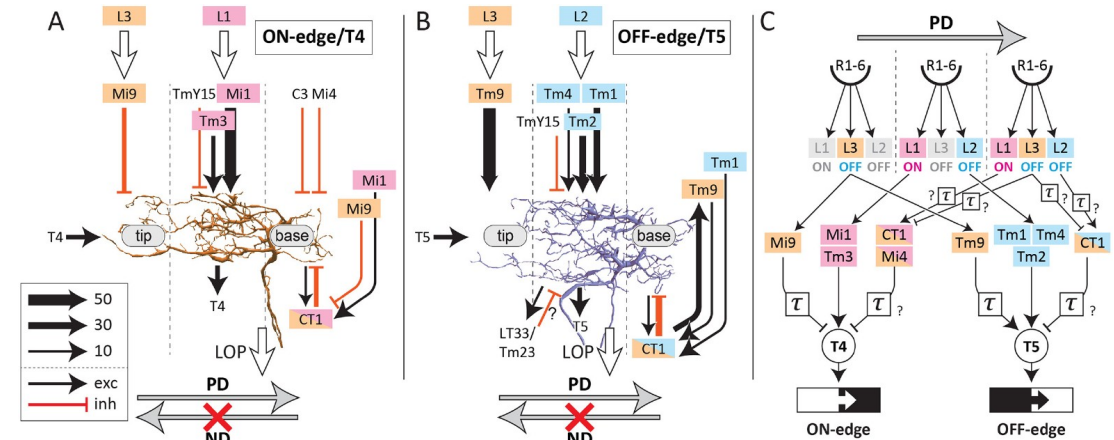
Schnell et al. (2012), Fig. 1(a, b). T4/T5 anatomy.

Bright and dark edges use parallel motion pathways

An **ON edge** increases brightness as it passes a visual location; an **OFF edge** decreases it. The fly processes these motion signals through parallel pathways: T4 primarily carries ON-edge motion and T5 primarily carries OFF-edge motion.

Serial electron microscopy places the T4 and T5 input networks in M10 and Lo1, respectively. T4 receives medulla inputs including Mi1 and Tm3; T5 receives inputs including Tm1, Tm2, Tm4 and Tm9. Each pathway samples spatially arranged signals rather than a single intensity measurement.

Haag and colleagues measured direction-dependent calcium responses in both pathways. Parallel organization refers to different input populations and edge preferences, not to completely isolated computations. Opposite-polarity influences and baseline signals can still contribute to a neuron preferentially driven by one edge sign.



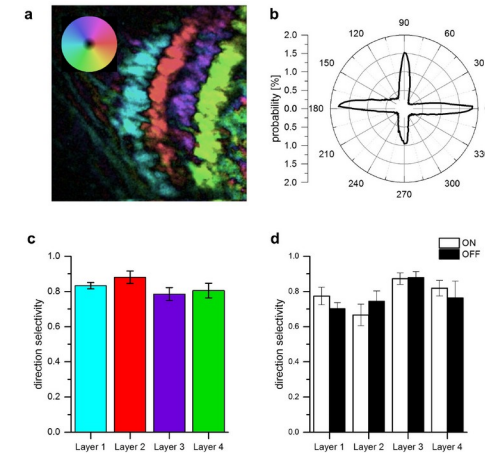
Shinomiya et al. (2019), Fig. 6. ON and OFF inputs converge onto distinct dendrites.

Four neuron subtypes encode four motion directions

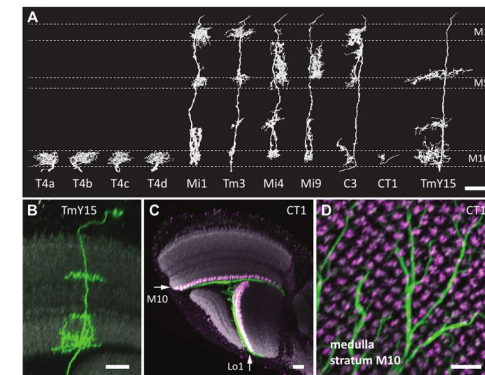
T4 and T5 each comprise four directional subtypes. Their axons terminate in four lobula plate layers representing front-to-back, back-to-front, upward and downward motion. A **preferred direction** produces the stronger response; its opposite is the **null direction**.

Haag and colleagues imaged calcium signals while gratings moved in different directions. Each layer contained neurons with a corresponding tuning preference, and both ON and OFF neurons contributed to the same layer-specific direction representation.

The layers organize motion direction rather than contrast polarity: matching T4 and T5 subtypes converge despite their different edge preferences. Takemura's electron-microscopic reconstructions distinguish the T4 subtypes anatomically. Combining reconstruction with functional tuning links dendritic organization to a direction-specific output channel.



Haag et al. (2017), Fig. 4. Directional tuning across four output layers.



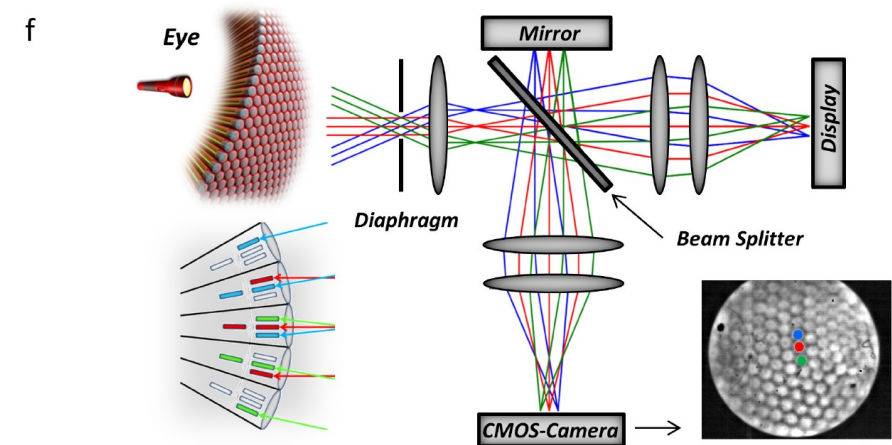
Takemura et al. (2017), Fig. 2(A–D). Reconstructed T4 subtypes and optic-lobe anatomy.

Calcium imaging resolves individual receptive fields

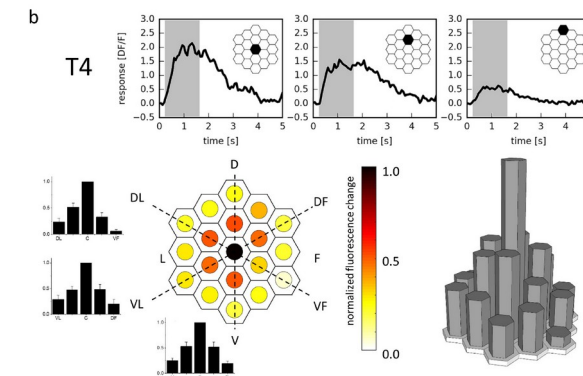
A **receptive field** is the set of visual positions influencing a neuron's response. Haag and colleagues stimulated individual **ommatidia**, the repeated optical units of the compound eye, while imaging T4/T5 terminals in the lobula plate.

A custom optical system directed light to selected positions among 716 points covering approximately 60° by 60°. **GCaMP6m**, a genetically encoded fluorescent calcium indicator, reported activity as $\Delta F/F$, the fluorescence change relative to its baseline.

T4 neurons responded most strongly near the receptive-field center and less strongly at neighboring positions. Mapping isolated local responses established a reference for sequence experiments. Calcium fluorescence reports a consequence of cellular activity; it is not a direct measurement of membrane voltage or synaptic current.



Haag et al. (2016), Fig. 1(f). Single-ommatidium stimulation optics.



Haag et al. (2016), Fig. 2(b). Measured T4 receptive field and local calcium responses.

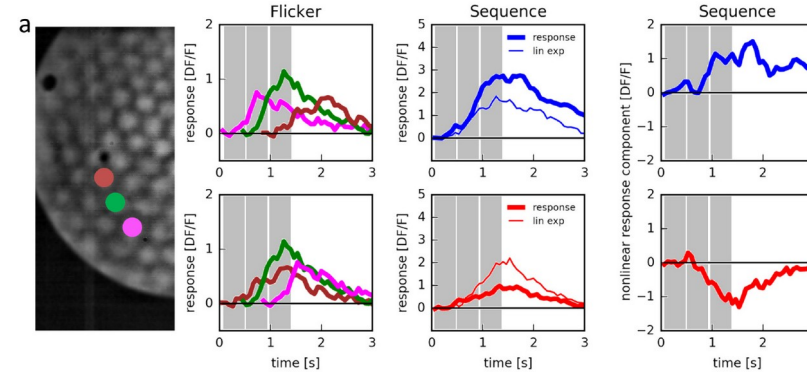
T4 responses depend on the order of local stimulation

A **linear prediction** adds the responses to separately presented inputs. Haag and colleagues first measured flashes at three neighboring locations, then compared their sum with the response when the same locations were stimulated in a motion sequence.

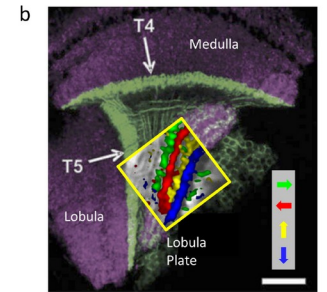
Preferred-direction sequences produced a measured T4 calcium response larger than the linear prediction.

Reversing the order produced a smaller response than predicted, despite using the same spatial positions. The order-dependent difference is a **nonlinear interaction**, because addition alone cannot reproduce it.

Subtracting the linear prediction from the measured response separated enhancement from suppression. The experiment locates direction-dependent computation at or before T4 output, but calcium imaging alone cannot identify the exact synaptic conductance generating the nonlinear component.



Haag et al. (2016), Fig. 3(a). Single-location controls and sequential T4 responses.



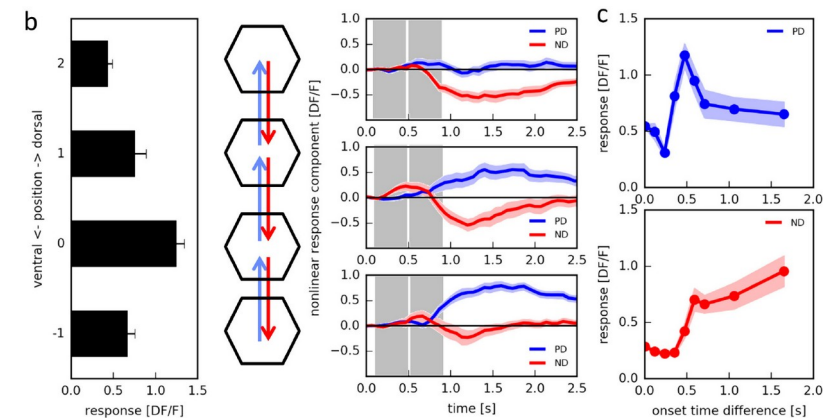
Haag et al. (2016), Fig. 1(b). T4/T5 anatomical compartments.

Preferred-direction stimulation enhances responses

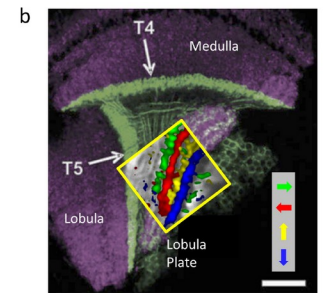
Preferred-direction enhancement means that a spatial sequence produces more activity than expected from adding its individual inputs. Haag and colleagues varied which positions were paired and the interval between them while monitoring T4 calcium signals.

Enhancement depended on the position of the preceding stimulus relative to the receptive-field center. Its largest measured contribution occurred at an onset separation near 0.1 s; changing the interval weakened the effect. The directional interaction therefore has both spatial and temporal structure.

A delayed excitatory interaction is consistent with a Reichardt-like enhancement mechanism. The response excess does not prove that two neurons perform a literal arithmetic multiplication, nor does the effective stimulus interval identify a single channel's opening time.



Haag et al. (2016), Fig. 3(b, c). Position and timing affect the nonlinear response.



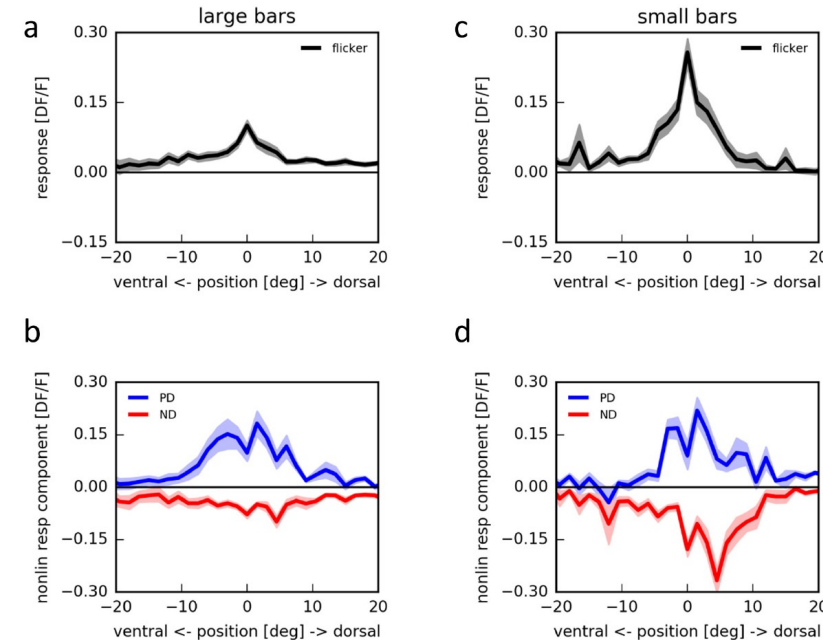
Haag et al. (2016), Fig. 1(b). T4/T5 anatomical compartments.

Null-direction stimulation suppresses responses

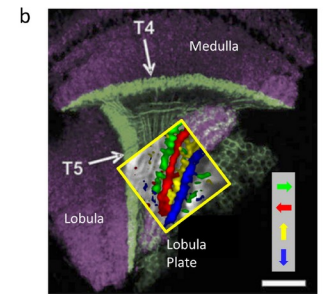
Null-direction suppression occurs when the reverse spatial sequence reduces a neuron's response below the linear prediction. Haag and colleagues measured this effect both with targeted ommatidium stimulation and with bars on an LED arena.

On the arena, isolated bars identified positions that excited T4. Sequential stimulation could nevertheless produce a negative nonlinear component for motion in the null direction. A response to one component therefore does not determine the response to its pairing with another.

Suppression is compatible with an inhibitory or divisive interaction, but the calcium response alone does not distinguish those biophysical possibilities. It complements preferred-direction enhancement: directional tuning can reflect both an increased response to one sequence and a reduced response to its reverse.



Haag et al. (2016), Fig. 4. Sequential bars produce below-linear null-direction responses.



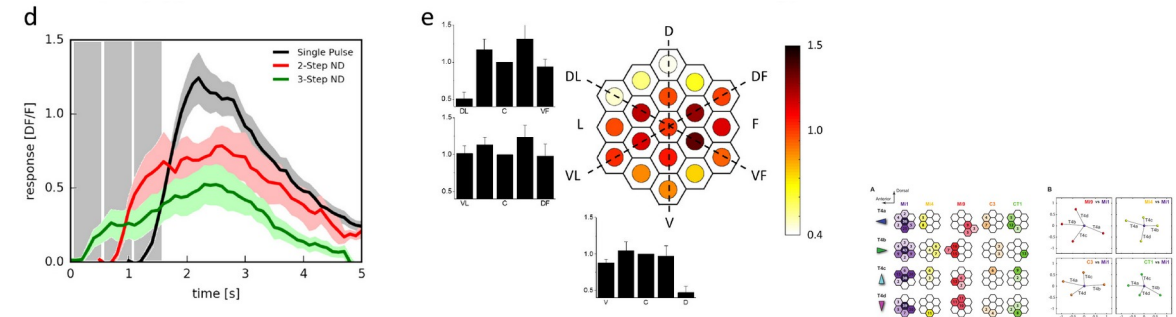
Haag et al. (2016), Fig. 1(b). T4/T5 anatomical compartments.

Enhancement and suppression occupy different positions

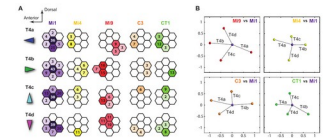
A T4 receptive field contains spatially different contributions to direction selectivity. Haag and colleagues mapped the nonlinear effects of stimulating the center together with neighboring ommatidia. Enhancement and suppression were concentrated on different sides of the center.

Takemura and colleagues reconstructed synaptic partners across neighboring medulla columns. Inputs from Mi1 and Tm3 were concentrated centrally, whereas other input types occupied displaced regions of the T4 dendrite. Spatial asymmetry is therefore present in both functional responses and anatomical connectivity.

The studies used different measurements and preparations. Correspondence between displaced inputs and nonlinear response regions constrains candidate mechanisms; it does not assign every physiological component to one identified synapse or directly measure transmission delays in the reconstructed neurons.



Haag et al. (2016), Fig. 3(d, e). Neighboring positions differ in their nonlinear influence.



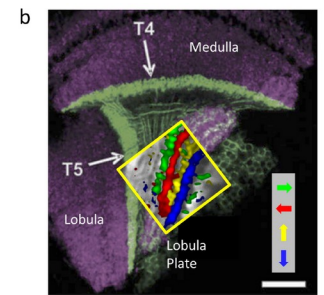
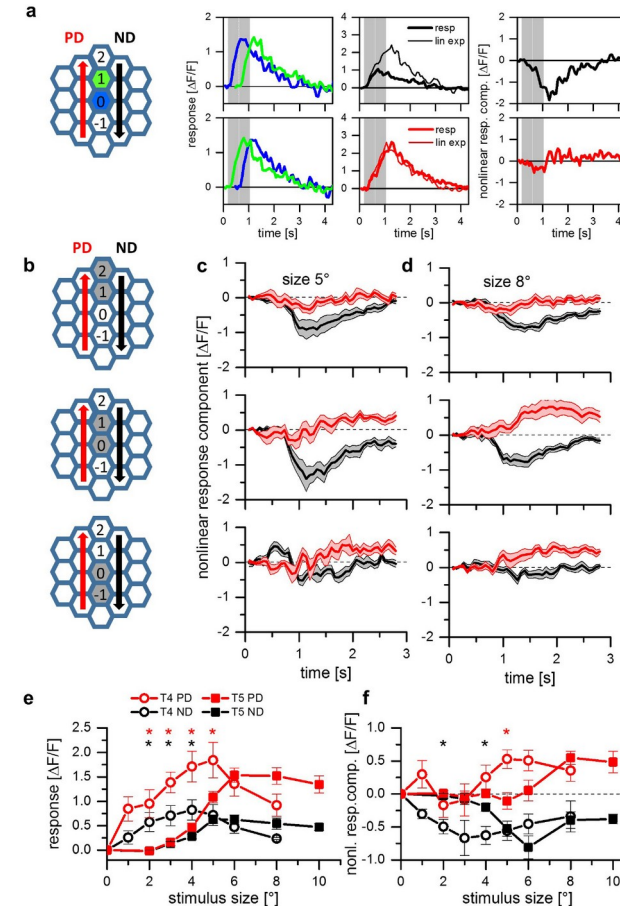
Takemura et al. (2017), Fig. 4(A, B). T4 input distributions across neighboring columns.

T5 combines the same complementary operations

T5 neurons preferentially respond to OFF-edge motion. Haag and colleagues stimulated neighboring visual columns with brightness decrements while imaging T5 terminal calcium, testing whether dark-edge motion required a different directional operation from T4.

A two-step sequence near the receptive-field center could produce both preferred-direction enhancement and null-direction suppression. Increasing the stimulus-spot diameter from 5° to 8° revealed preferred-direction enhancement alongside suppression.

The same classes of nonlinear operation can therefore occur in both ON and OFF pathways despite different input neurons and dendritic locations. A shared computation does not require identical receptive-field size, synaptic chemistry or temporal filtering in T4 and T5.



Haag et al. (2016), Fig. 1(b). T4/T5 anatomical compartments.

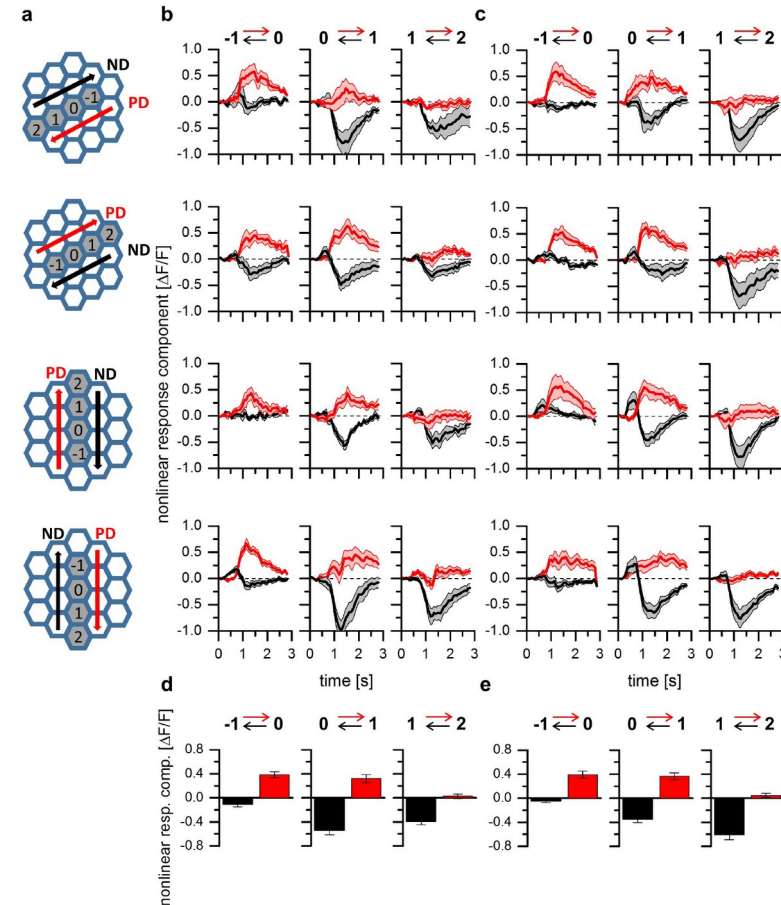
Haag et al. (2017), Fig. 2(a-f). OFF-cell sequences produce enhancement and suppression.

Complementary operations span all T4 and T5 subtypes

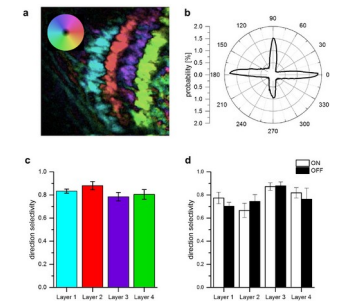
A mechanism measured in one directional subtype need not apply to all local motion neurons. Haag and colleagues compared T4 and T5 across all four lobula plate layers using two-step apparent-motion stimuli aligned with each subtype's preferred axis.

Preferred-direction enhancement occurred when the stimulus sequence traversed one side of the receptive field toward its center. Null-direction suppression occurred for the reverse sequence involving the complementary side. The comparison included both horizontal and vertical motion channels.

Enhancement and suppression therefore form a common functional organization across the tested ON and OFF subtypes. The subtype comparisons establish the generality of these operations under the experimental conditions; they do not establish that every input weight, channel complement or adaptation response is identical.



Haag et al. (2017), Fig. 3. Nonlinear responses across four ON and OFF subtypes.



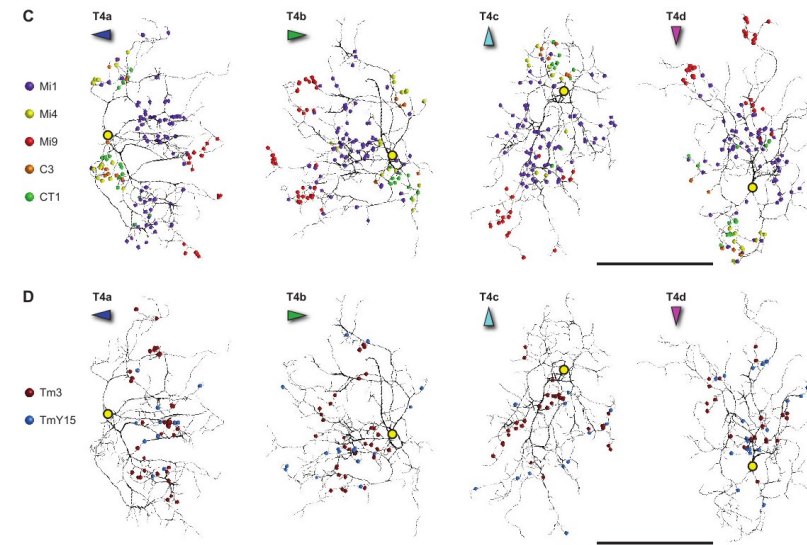
Haag et al. (2017), Fig. 4. Four direction layers.

Synaptic partners occupy different parts of T4 dendrites

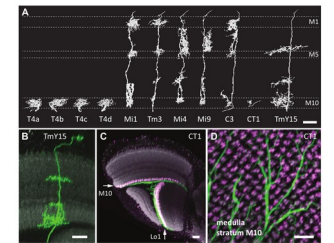
A **connectome** is a reconstruction of neurons and their synaptic connections. Takemura and colleagues traced inputs to four T4 subtypes in a focused-ion-beam scanning electron-microscopy volume, assigning contacts to identified presynaptic cell types.

Eight input classes accounted for more than 96% of the reconstructed T4 synapses. Their contacts were distributed unevenly: some concentrated along the dendritic shaft, some toward its tip, and others near the base. A dendrite therefore receives different combinations of signals at different locations.

Input position constrains which spatial signals can interact within T4. Synapse number measures anatomical connectivity, not physiological strength. The reconstruction does not directly establish receptor density, release probability or the response kinetics of an individual connection.



Takemura et al. (2017), Fig. 4(C, D). Synapse locations on four reconstructed T4 subtypes.



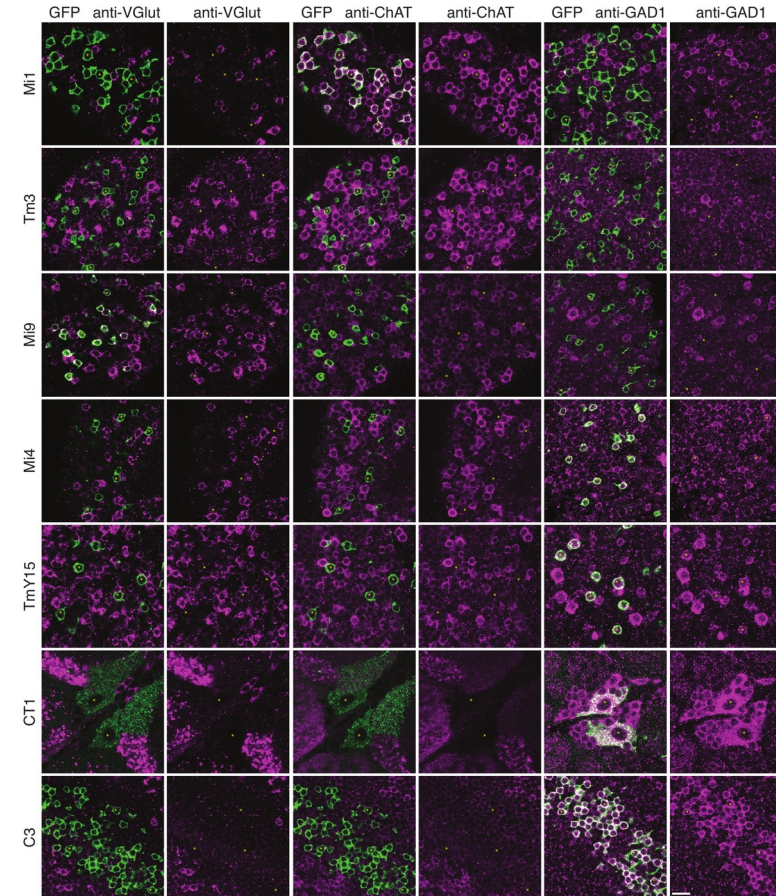
Takemura et al. (2017), Fig. 2(A–D). T4 inputs.

Central T4 inputs include Mi1 and Tm3

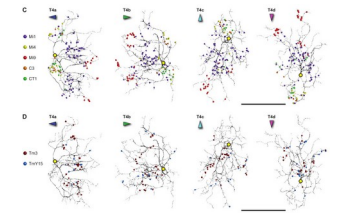
Mi1 and Tm3 are medulla interneurons relaying visual input to T4. Takemura and colleagues found that these two cell types supplied the largest portions of reconstructed T4 input, approximately 35% and 16% of its synaptic contacts, respectively.

Their synapses overlapped substantially along the dendritic shaft. Antibody labeling detected **choline acetyltransferase**, the enzyme that synthesizes acetylcholine, in Mi1 and Tm3. The combined anatomical and chemical evidence supports major cholinergic inputs to T4.

The expanded reconstruction did not reproduce a consistent Mi1–Tm3 spatial offset proposed from an earlier volume. Strong central input alone therefore does not identify the complete Reichardt detector. Additional displaced inputs and their functional time courses must be considered when interpreting directional computation.



Takemura et al. (2017), Fig. 5. Transmitter-marker staining of T4 inputs.



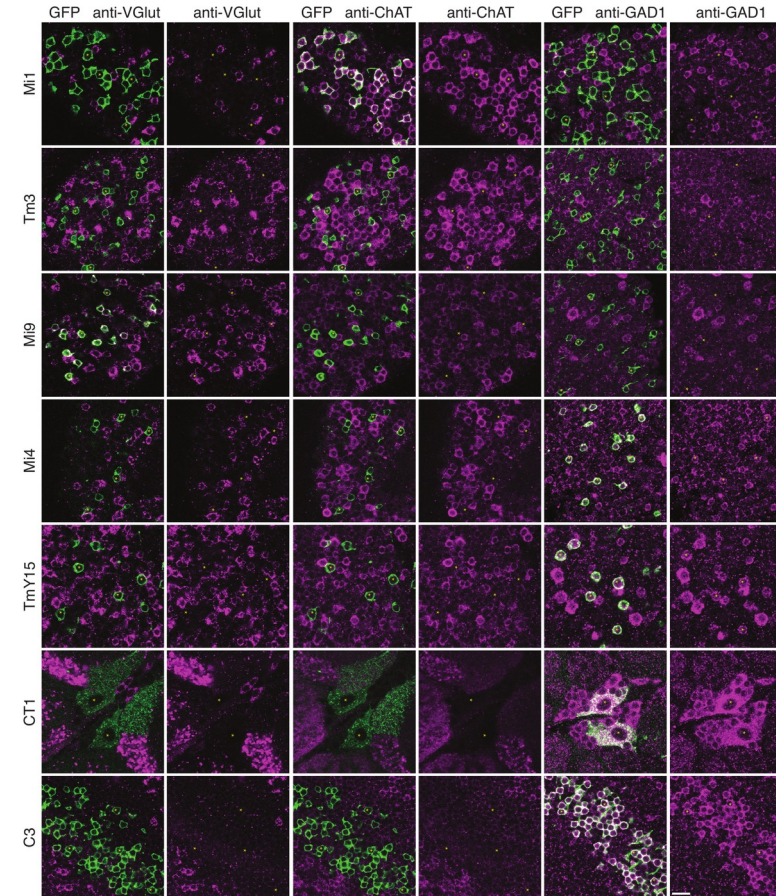
Takemura et al. (2017), Fig. 4(C, D). T4 synapses.

T4 also receives inhibitory and glutamatergic inputs

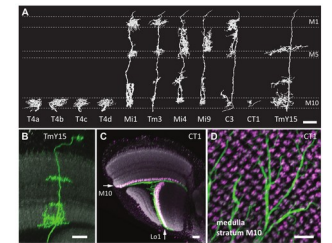
Mi4 and Mi9 contribute spatially displaced inputs to T4. Takemura and colleagues combined synapse reconstruction with immunolabeling, staining identified cell bodies for acetylcholine, glutamate and GABA-related markers rather than assigning chemistry from morphology alone.

Mi4 labeled for **GAD1**, glutamic acid decarboxylase, which synthesizes the neurotransmitter **GABA**. Mi9 labeled for **VGlut**, a transporter that packages glutamate into synaptic vesicles. These markers support GABAergic and glutamatergic input identities, respectively.

Glutamate is not necessarily excitatory in the fly. T4/T5 transcriptomics detected **GluCl α** , a glutamate-gated chloride channel associated with hyperpolarizing responses. Input chemistry, receptor identity and membrane conditions jointly determine response sign; a transmitter marker alone does not measure the current at a T4 synapse.



Takemura et al. (2017), Fig. 5. Transmitter markers in identified T4 input neurons.



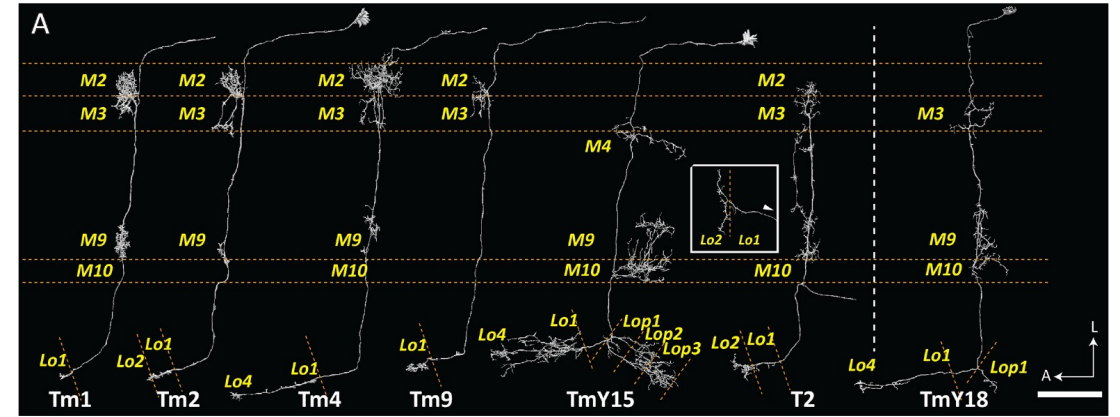
Takemura et al. (2017), Fig. 2(A–D). T4 inputs.

OFF-pathway inputs converge on T5 dendrites

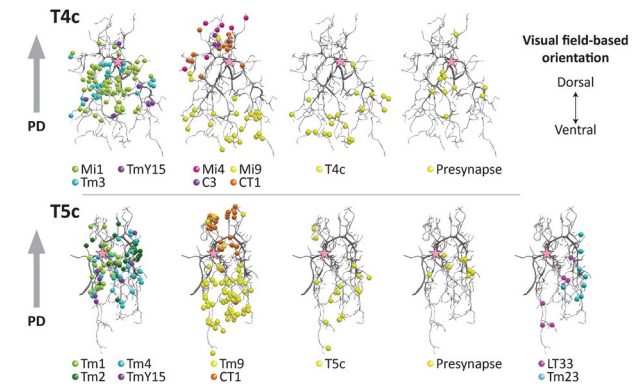
The **transmedullary** neurons Tm1, Tm2, Tm4 and Tm9 carry signals from the medulla to the lobula. Shinomiya and colleagues reconstructed their connections onto T5 dendrites in Lo1, linking the OFF pathway's relay cells to local motion computation.

Input locations were organized across each T5 dendrite. Tm1, Tm2 and Tm4 contributed centrally, while Tm9 supplied contacts displaced toward the dendritic tip. Other partners, including CT1, contributed near different dendritic regions.

These spatial relationships resemble the differentiated input zones of T4 while using a different set of relay neurons. Electron microscopy establishes connection geometry. The temporal ordering of those input signals and the nonlinear interaction producing an OFF-motion response require separate functional evidence.



Shinomiya et al. (2019), Fig. 3(A). Reconstructed OFF-pathway input neurons.



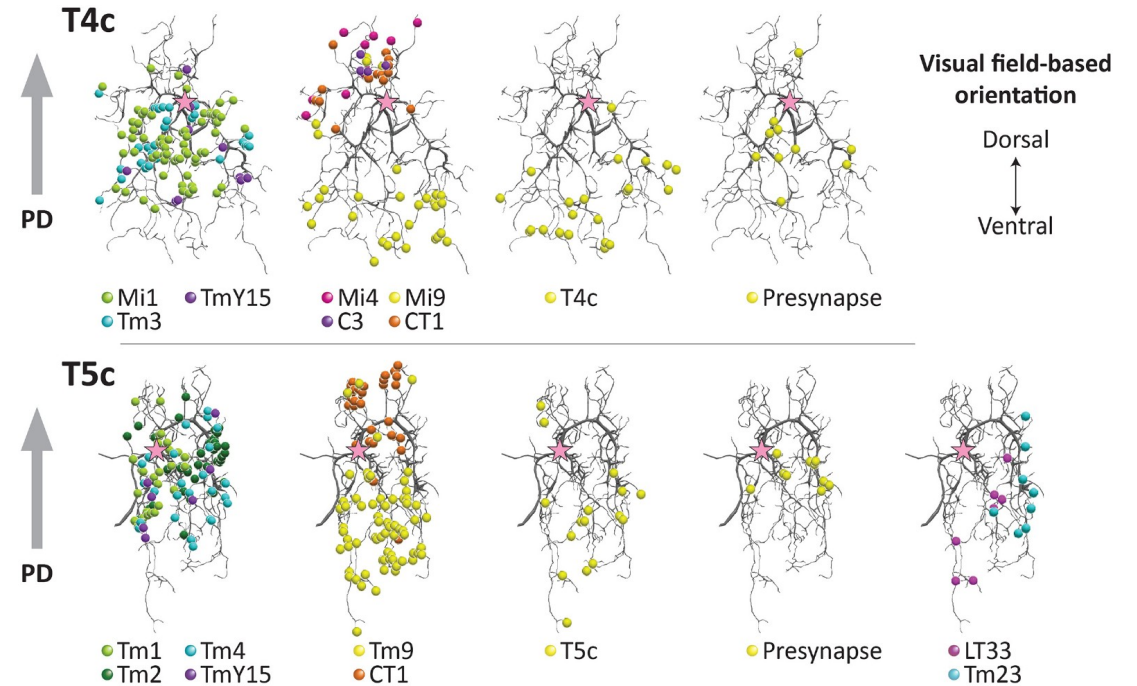
Shinomiya et al. (2019), Fig. 4. Synaptic distributions on T4 and T5 dendrites.

ON and OFF circuits share a spatial organization

A **presynaptic** neuron supplies a synaptic output, whereas the **postsynaptic** neuron receives it. Shinomiya and colleagues mapped input and output contacts along reconstructed T4 and T5 dendrites, distinguishing their partners and the direction of transmission.

Both pathways combined central contacts, displaced tip contacts and base-associated contacts. T4 received inputs including Mi1, Tm3, Mi9 and inhibitory partners; T5 combined Tm1, Tm2, Tm4, Tm9 and CT1. Four directional subtypes rotated the effective arrangement relative to visual space.

The shared pattern is a repeated organizational principle, not identical wiring. Additional partners and local outputs differ between the pathways. A circuit comparison therefore supports common computational constraints while preserving pathway-specific opportunities for timing, feedback and contrast dependence.



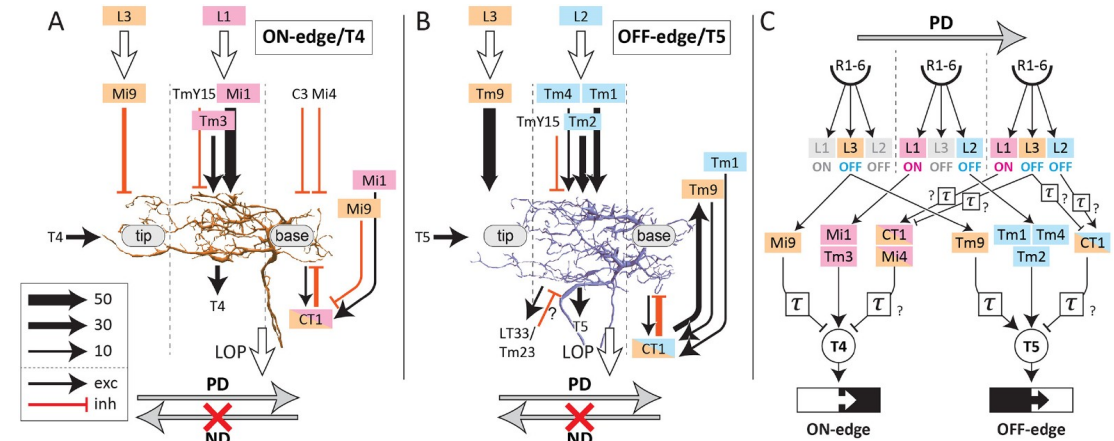
Shinomiya et al. (2019), Fig. 4. ON and OFF synapses occupy comparable dendritic zones.

Connectomes constrain circuits without measuring time

A spatially ordered input network can support delayed enhancement and suppression, but an anatomical contact does not specify a temporal filter. Synaptic counts and dendritic geometry constrain candidate motion computations without recording the activity of their input channels.

Takemura and Shinomiya reconstructed multiple relay routes reaching T4 or T5, including paths with different numbers of intervening synapses. Their published circuit accounts distinguish known connectivity from proposed excitatory, inhibitory and timing contributions.

A longer anatomical route need not have a uniquely determined delay, and a large contact count need not imply a fixed response weight. Physiology, receptor measurements and selective perturbation must establish those functional properties. The Reichardt algorithm and connectome can be consistent without identifying a single two-neuron implementation.



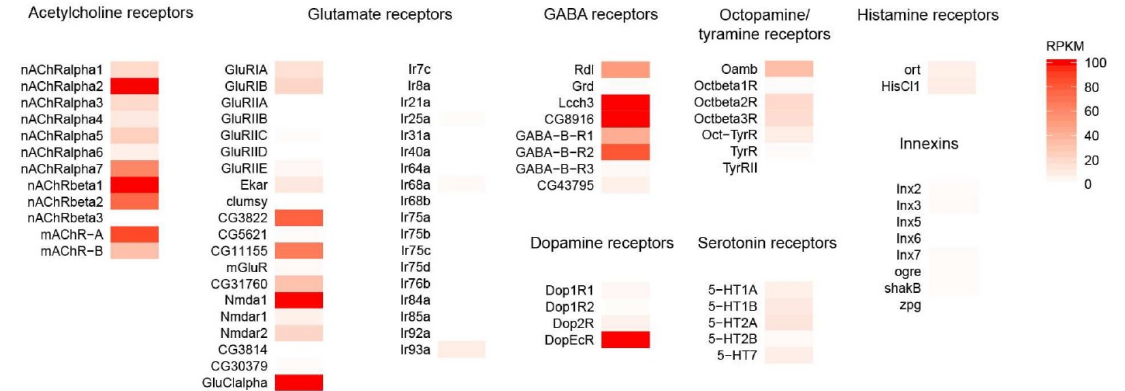
Shinomiya et al. (2019), Fig. 6. Published circuit account distinguishes measured and proposed links.

T4 and T5 express multiple receptor classes

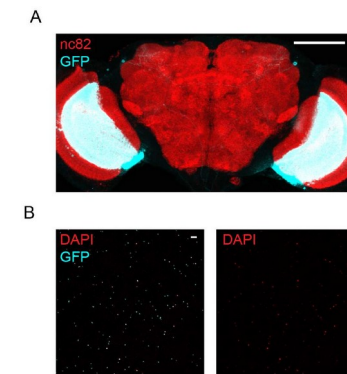
A **receptor** is a protein through which a neuron detects a signal such as a neurotransmitter. An **ionotropic** receptor directly controls an ion channel; a **metabotropic** receptor acts through intracellular signaling. Either class can alter a neuron's integration of input.

Pankova and Borst isolated genetically labeled T4/T5 nuclei and used **RNA sequencing** to measure expressed transcripts. They detected nicotinic and muscarinic acetylcholine receptor transcripts, GABA receptor transcripts and the glutamate-gated chloride channel GluCl α .

The published heat map reports **RPKM**, reads per kilobase of transcript per million mapped reads, on a 0–100 color scale. The assay pooled T4 and T5 and measured RNA. It did not establish protein abundance, receptor location or functional responses separately in each cell class.



Pankova & Borst (2016), Fig. 3. Receptor transcripts quantified on the original expression scale.



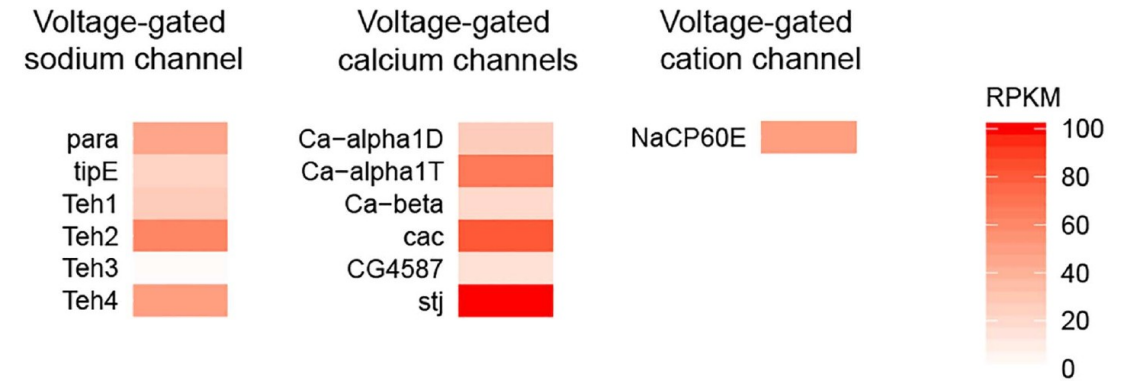
Pankova & Borst (2016), Fig. 2(A, B). Labeled T4/T5 neurons and isolated nuclei.

Sodium and calcium channels shape membrane signaling

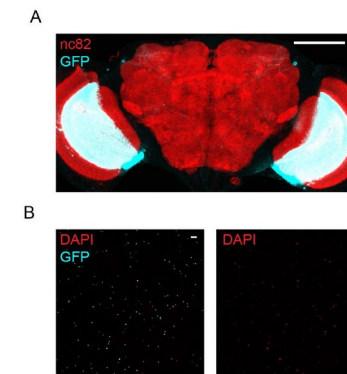
A **voltage-gated ion channel** changes its opening probability with membrane potential, the electrical potential across the cell membrane. Such channels can transform synaptic input into different voltage and calcium responses instead of transmitting a passive copy of the input.

Pankova and Borst detected the sodium-channel gene **para**, four of its five identified auxiliary subunits, and transcripts for the identified voltage-gated calcium-channel subunits. They also detected NaCP60E, a cation channel permeable to sodium and calcium.

Channel expression provides molecular candidates for active processing and calcium entry in T4/T5. The transcriptome does not establish that these neurons fire sodium action potentials, identify the channels carrying a recorded calcium signal, or prove that one channel implements the detector's effective temporal delay.



Pankova & Borst (2016), Fig. 4. Voltage-gated channel transcripts in pooled T4/T5 neurons.



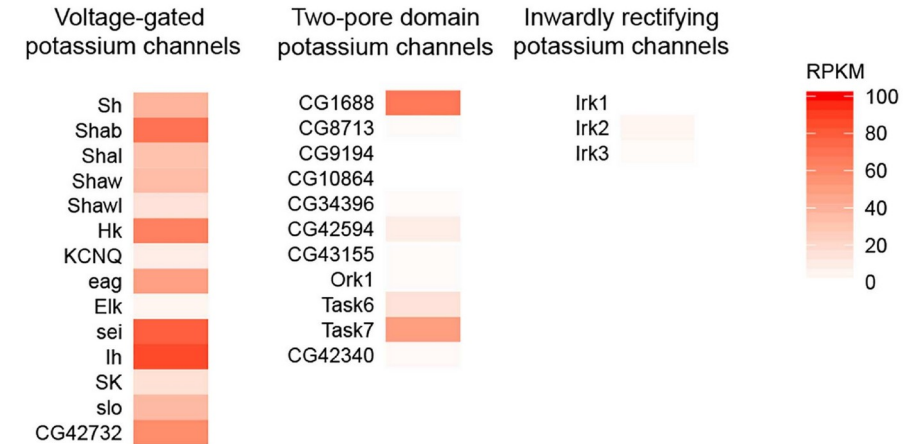
Pankova & Borst (2016), Fig. 2(A, B). Labeled T4/T5 neurons and isolated nuclei.

Potassium channels can regulate temporal integration

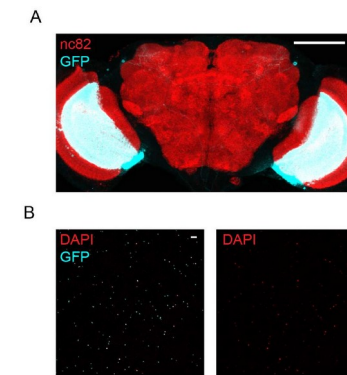
Potassium channels conduct potassium ions and can influence how membrane voltage evolves after excitation. **Temporal integration** is the combination of inputs over time; channel kinetics can alter the interval over which previous input affects a neuron's response.

Pankova and Borst detected transcripts for multiple voltage-gated and two-pore-domain potassium channels in pooled T4/T5 nuclei. Shaker, Shal and SK were among the strongly expressed candidates; inwardly rectifying potassium-channel transcripts were not detected in their assay.

The authors proposed that active membrane conductances could contribute to delays and nonlinear interactions within motion detectors. This is a mechanistic possibility supported by molecular capacity, not a demonstration that a named potassium channel generates preferred-direction enhancement or null-direction suppression. Channel-specific causal experiments are required.



Pankova & Borst (2016), Fig. 5. Potassium-channel transcripts on the original RPKM scale.



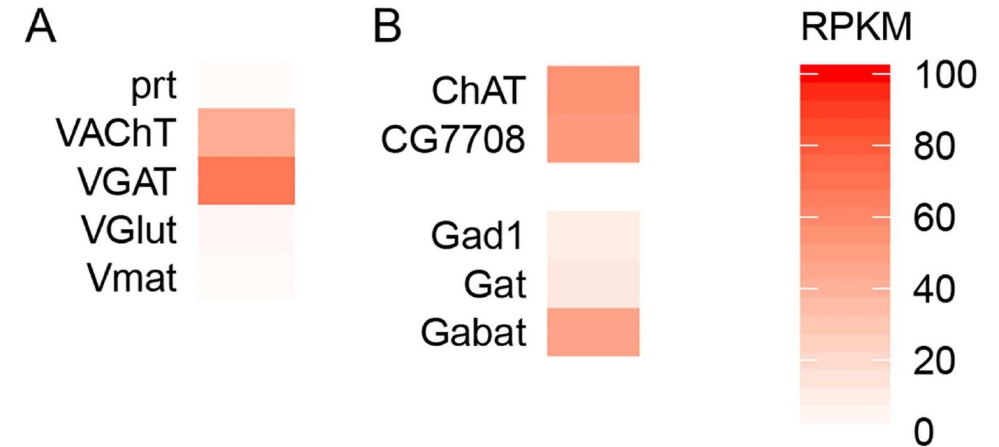
Pankova & Borst (2016), Fig. 2(A, B). Labeled T4/T5 neurons and isolated nuclei.

T4 and T5 have cholinergic transmitter machinery

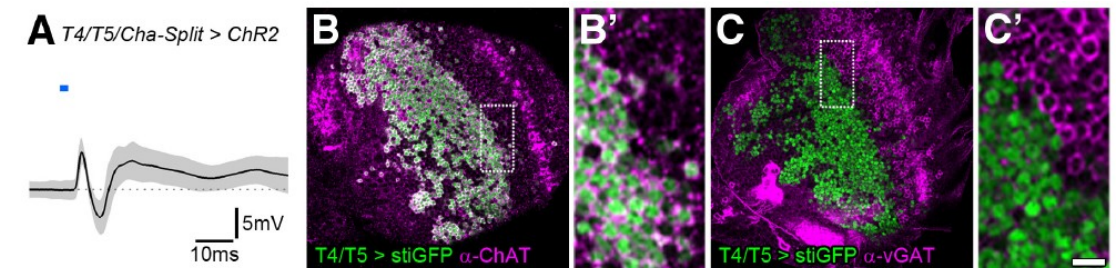
A neuron's **transmitter phenotype** describes the chemical signaling machinery it uses. **Choline acetyltransferase** synthesizes acetylcholine, and the **vesicular acetylcholine transporter** packages it for synaptic release. Pankova and Borst detected transcripts for both in T4/T5.

Mauss and colleagues combined cell labeling with optogenetic stimulation and recordings from downstream lobula plate neurons. T4/T5 terminals labeled with the acetylcholine-synthesis marker, and selective activation of cholinergic T4/T5 produced rapid excitation followed by inhibition.

The transcriptome also detected vesicular GABA-transporter expression, creating a molecular question about additional signaling. Expression alone does not establish functional GABA release. The physiological and pharmacological results support a primarily cholinergic output with an indirect inhibitory route rather than direct inhibitory transmission from every T4/T5 terminal.



Pankova & Borst (2016), Fig. 6. Transcripts for transmitter packaging and synthesis.



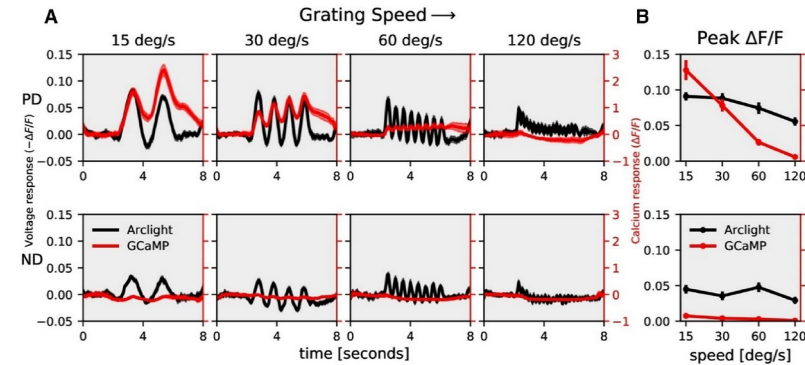
Mauss et al. (2014), Fig. 3. Physiology and transmitter-marker staining in T4/T5.

Calcium can sharpen voltage-based direction selectivity

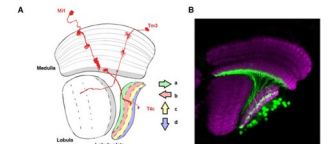
ArcLight is a fluorescent voltage indicator; **GCaMP6f** reports intracellular calcium. Mishra and colleagues expressed them in T4c neurons, whose preferred direction is upward, and compared dendritic responses to the same visual stimulus set.

Gratings had wavelength 30° and moved at 15, 30, 60 or $120^\circ/\text{s}$. Voltage responses retained substantial null-direction activity, whereas calcium responses were more strongly biased toward the preferred direction. The difference was particularly pronounced at slower speeds.

The voltage-to-calcium transformation therefore contributes to measured direction selectivity rather than simply copying membrane voltage. Calcium-based estimates of the circuit's selectivity can exceed voltage-based estimates. The comparison used different labeled preparations for the two indicators, so it does not amount to simultaneous voltage and calcium recording from one neuron.



Mishra et al. (2023), Fig. 2(A, B). ArcLight and GCaMP motion responses and their speed dependence.



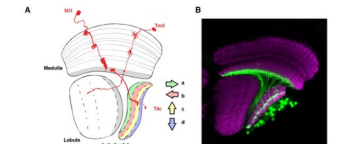
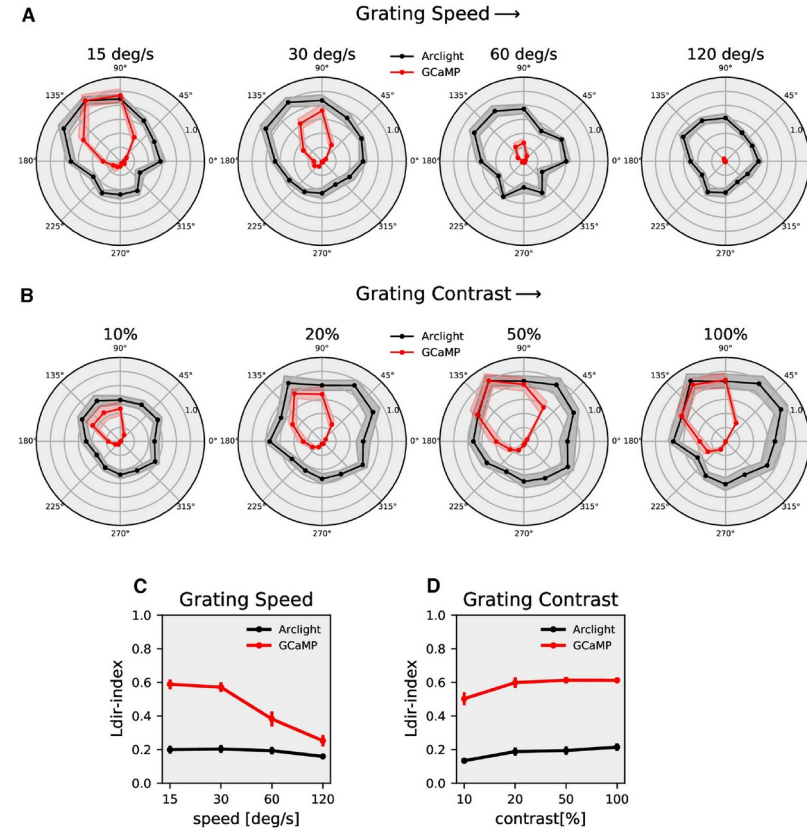
Mishra et al. (2023), Fig. 1(A, B). T4c anatomy.

Calcium signals differ from synaptic computations

A neuron's **directional tuning** is its response across motion directions. Mishra and colleagues measured T4c responses to gratings moving in 12 directions, across four speeds and contrasts from 10% to 100%, using voltage and calcium indicators.

Calcium tuning was narrower and more direction-selective than voltage tuning under the tested conditions. Both signals changed with speed and contrast, but their relative sensitivity differed. The readout is therefore part of the experimentally observed transformation from visual input to neural output.

The authors compared published filtering and nonlinear-transform models with the measurements. A successful model describes the relationship between indicators; it does not identify a unique channel or synaptic receptor. Synaptic direction computation and the later voltage-to-calcium transformation are distinct contributions that can coexist within T4.



Mishra et al. (2023), Fig. 1(A, B). T4c anatomy.

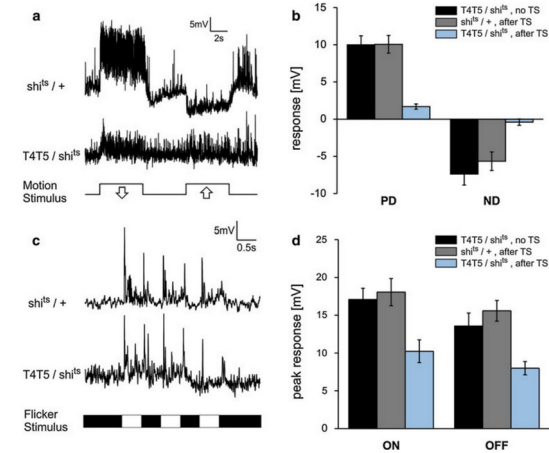
Mishra et al. (2023), Fig. 4. Voltage and calcium directional tuning across speed and contrast.

Lobula plate neurons pool many local motion signals

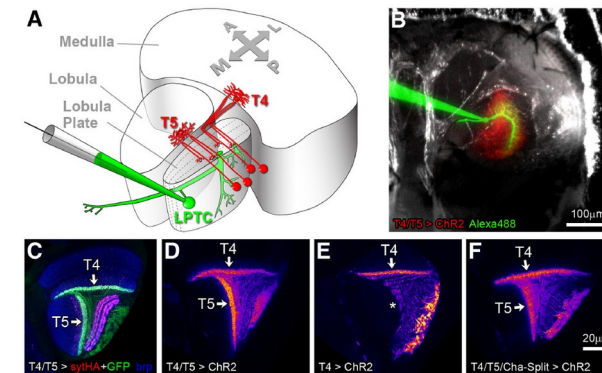
Lobula plate tangential cells, or LPTCs, extend dendrites across many local motion channels. **Spatial pooling** combines activity from multiple visual locations, allowing these neurons to represent broad optic-flow patterns rather than only one neighboring pair of inputs.

Schnell and colleagues recorded horizontal-system and vertical-system cells, which favor horizontal and vertical flow components. Blocking T4/T5 output removed motion-dependent voltage responses while leaving responses to brightness flicker. The wide-field directional signal therefore depends on the local detector population.

Mauss and colleagues labeled T4/T5 terminals together with recorded LPTCs, relating the input population to its downstream targets. Pooling explains a change in spatial scale; its anatomical presence does not specify the complete transformation from lobula plate activity to steering torque or walking turns.



Schnell et al. (2012), Fig. 2. Motion responses disappear while flicker responses persist.



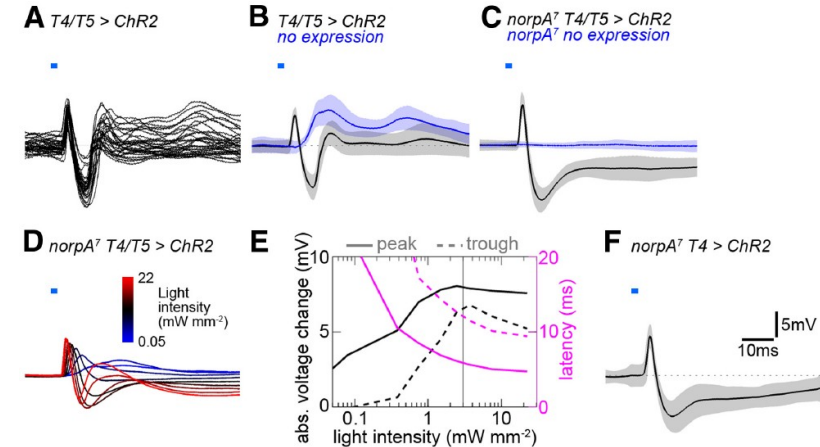
Mauss et al. (2014), Fig. 1(A-F). Local motion neurons contact lobula plate tangential cells.

T4 and T5 activation excites tangential cells

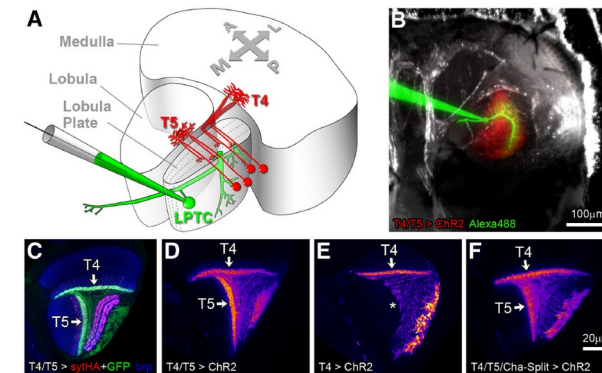
Optogenetics uses a light-sensitive protein to manipulate selected neurons. Mauss and colleagues expressed **channelrhodopsin-2**, a light-gated cation channel, in T4/T5 and recorded downstream LPTC membrane voltage during 2-ms blue-light pulses.

Selective activation produced a fast depolarizing peak at approximately 5.4 ms followed by an inhibitory trough near 11 ms. **Depolarization** makes membrane potential less negative; **hyperpolarization** makes it more negative. These response components have different timing despite the same initiating stimulus.

Flies lacking channelrhodopsin provided a light-stimulation control, and visually blind *norpA* preparations reduced contamination from ordinary retinal responses. The experiments establish a rapid excitatory route from T4/T5 to LPTCs; the later inhibitory component requires an additional explanation rather than a second assumed direct transmitter action.



Mauss et al. (2014), Fig. 2. Brief optogenetic pulses evoke fast excitation and delayed inhibition.



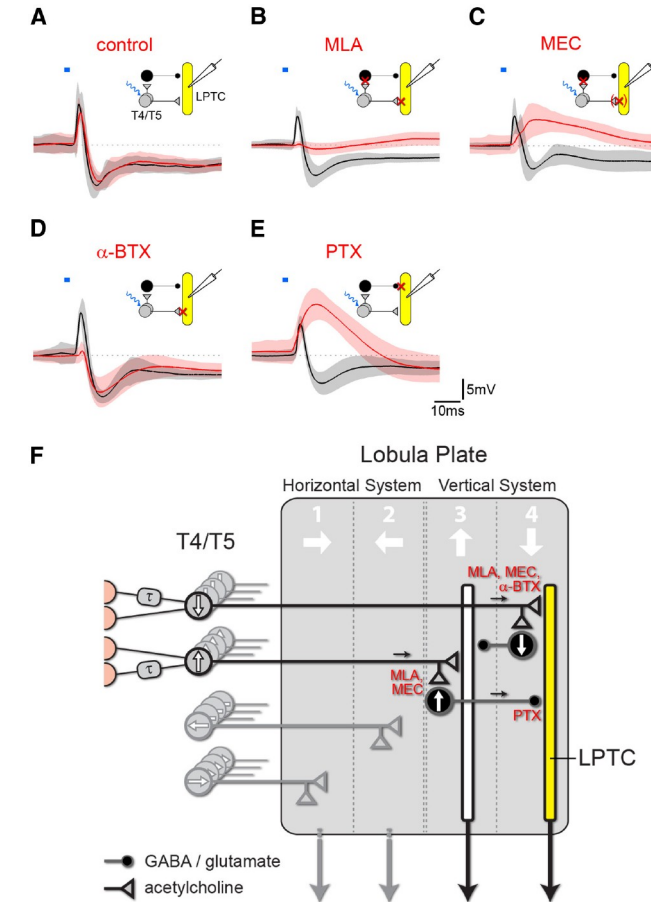
Mauss et al. (2014), Fig. 1(A–F). Labeled T4/T5 inputs and the recorded tangential cell.

Feedforward inhibition supplies opposite-direction signals

Feedforward inhibition occurs when an upstream neuron activates an intervening inhibitory neuron that suppresses a downstream target. Mauss and colleagues combined T4/T5 optogenetic stimulation with antagonists, drugs that reduce signaling through particular receptor classes.

Blocking nicotinic acetylcholine receptors with methyllycaconitine strongly reduced both response components. α -Bungarotoxin preferentially reduced fast excitation, whereas picrotoxin removed inhibition and increased excitation. The different profiles support direct cholinergic excitation plus an indirectly recruited inhibitory route.

Picrotoxin at 25 μ M also reduced responses to pressure-applied GABA and glutamate, so it did not uniquely identify GABA transmission. The intermediate inhibitory cell was not identified in this study. Receptor pharmacology constrains circuit organization without proving one exclusive inhibitory transmitter or a complete cell-by-cell pathway.



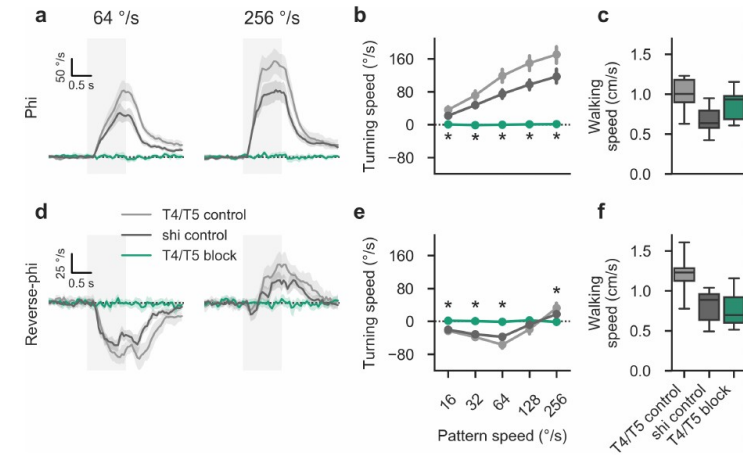
Mauss et al. (2014), Fig. 5(A–F). Pharmacology and the published feedforward circuit.

Reverse-phi turning requires T4 and T5

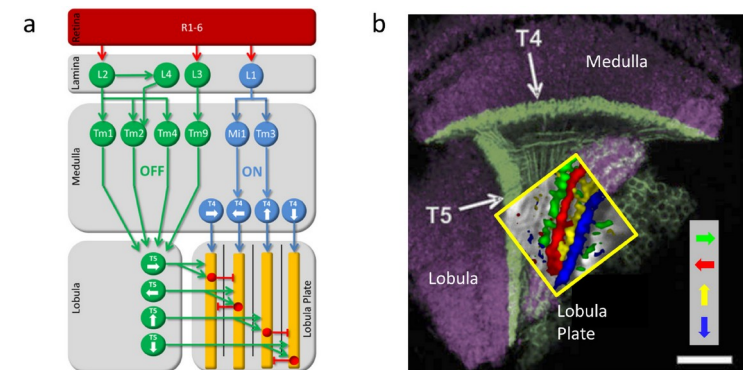
A motion illusion can use the same early circuit as an ordinary stabilizing response. Leonhardt and colleagues suppressed T4/T5 synaptic output and measured turning during phi and reverse-phi stimulation in tethered walking flies.

Control flies turned with ordinary motion and against slower reverse-phi motion. T4/T5-blocked flies lost both responses, including the high-velocity re-inversion of reverse-phi turning. Genetic controls retained directional responses under the same test conditions.

Forward walking speed remained comparable across the tested genotypes. The behavioral deficit therefore was not explained by an inability to walk. T4/T5 are necessary for these directional responses, but joint silencing does not establish which pathway or synaptic interaction generates each component of the illusion.



Leonhardt et al. (2017), Fig. 4. T4/T5 silencing removes turning but preserves walking.



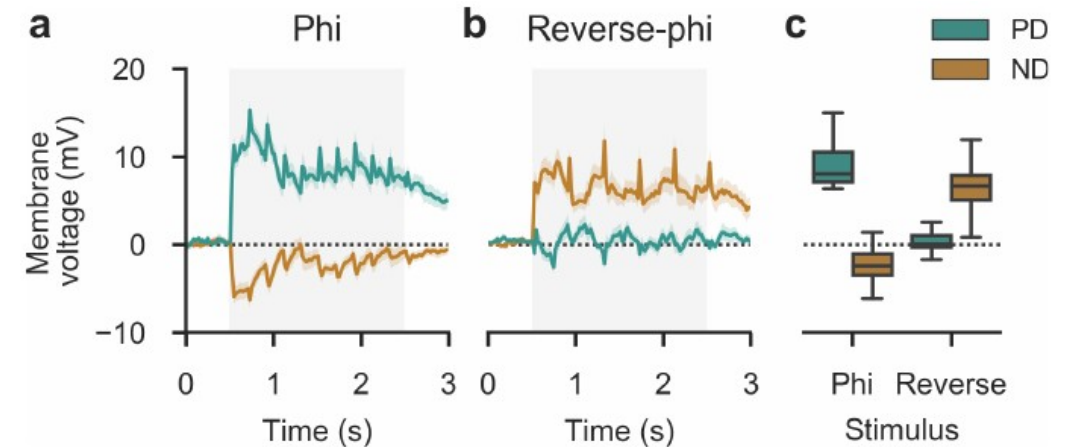
Haag et al. (2016), Fig. 1(a, b). Local ON and OFF pathway anatomy.

Tangential cells can encode an illusory direction

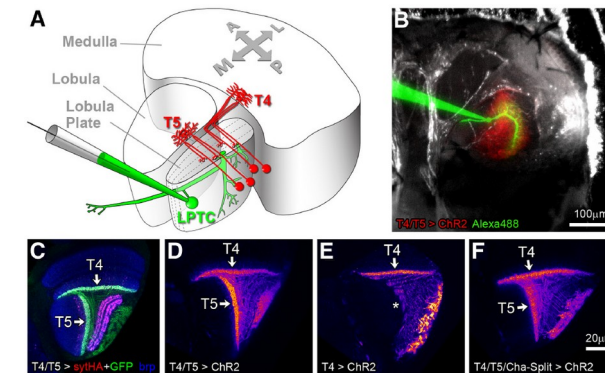
A **patch-clamp recording** measures electrical activity through an electrode contacting a neuronal membrane. Leonhardt and colleagues recorded tangential-cell voltage during phi and reverse-phi gratings with wavelength 45° , speed $128^\circ/\text{s}$ and step width 6° .

Ordinary preferred-direction motion depolarized the cells, and ordinary null-direction motion hyperpolarized them. Reverse-phi displaced in the null direction instead produced depolarization, though weaker than the ordinary preferred-direction response. Reverse-phi displaced in the preferred direction produced little detectable voltage deflection.

The neuronal response therefore partially inverted with the illusion rather than simply reproducing every behavioral response symmetrically. Direct voltage recordings distinguish hyperpolarization from a fluorescence signal near zero. Agreement between neural and behavioral direction supports an early sensory contribution without proving a complete one-to-one mapping onto turning.



Leonhardt et al. (2017), Fig. 5. Patch-clamp responses to phi and reverse-phi motion.



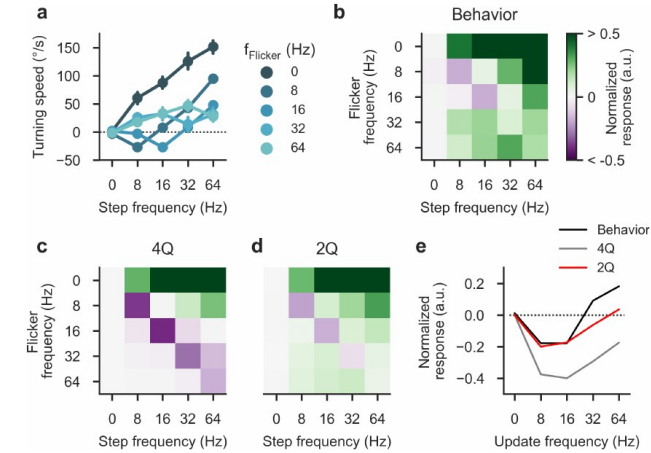
Mauss et al. (2014), Fig. 1(A–F). Lobula plate inputs and tangential-cell anatomy.

Bright and dark motion pathways can interact

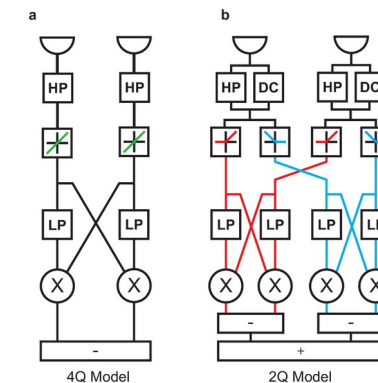
Rectification preferentially retains one sign of an input signal. A motion model that completely rectifies bright and dark signals into independent channels can lose the cross-polarity interactions required for reverse-phi responses. Separate ON and OFF pathways therefore need a more precise computational description.

Leonhardt and colleagues varied flicker frequency independently of spatial-step frequency. Behavioral responses changed with both, constraining explanations based only on physical displacement. A published two-quadrant model including a constant-luminance component captured important features of the measured reversal.

The model offers an account of effective pathway interactions, not a uniquely established receptor mechanism. Clark's polarity tests and Leonhardt's dissociated stimuli both require sensitivity beyond two isolated, perfectly rectified streams. Anatomical ON/OFF separation remains compatible with functional influence from both contrast signs.



Leonhardt et al. (2017), Fig. 9. Measured responses and published detector predictions.



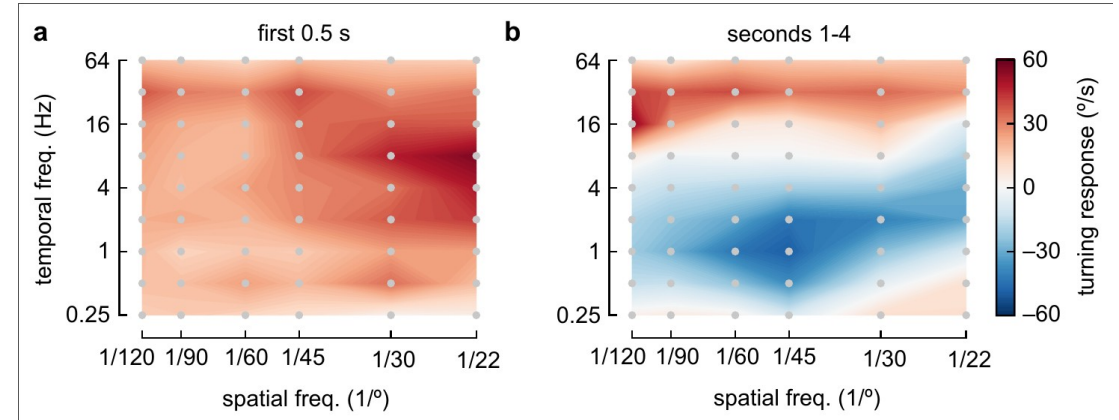
Leonhardt et al. (2017), Fig. 6. Published four- and two-quadrant computational accounts.

Sustained high contrast can reverse optomotor turning

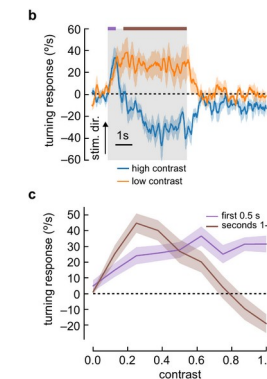
Reverse-phi changes stimulus contrast sign between frames. **Anti-directional turning** in Mano's experiments occurred during continuously drifting sinusoidal gratings without such contrast reversals, so the two behavioral reversals have different stimulus definitions.

With wavelength 60° , temporal frequency 1 Hz and contrast 1.0, flies initially turned with the pattern and then against it after approximately 1–2 s. At contrast 0.25, the turn remained aligned with the stimulus. The pattern's physical motion did not reverse.

Spatial and temporal tuning also differed between the initial 0.5-s response and the later 1–4-s response. A fixed mapping from local motion direction to stabilizing turns cannot explain the full time course. Downstream dynamics or adaptation are required, but the behavioral timing alone does not identify their circuit location.



Mano et al. (2023), Fig. 2(a, b). Early and late turning differ across spatial and temporal frequency.



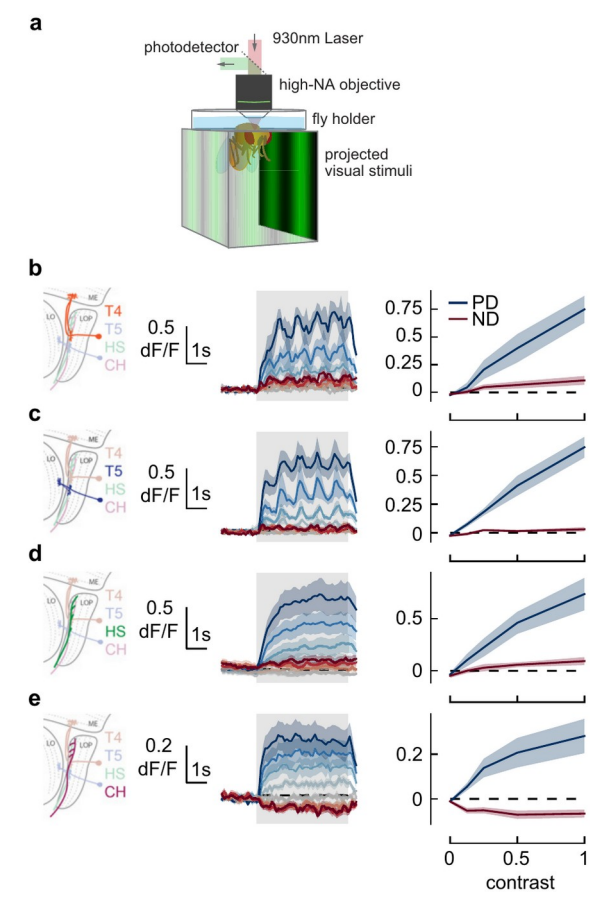
Mano et al. (2023), Fig. 1(b, c). Turning changes sign during an unchanged high-contrast stimulus.

Behavioral reversals need not originate in T4 or T5

A neuron can be necessary for a behavior without generating every temporal transformation within that behavior. Mano and colleagues silenced T4/T5 and eliminated both ordinary and delayed anti-directional turning during the tested gratings.

Calcium imaging of T4, T5 and several lobula plate cell classes did not reveal a reversal or decay matching the behavioral switch. Preferred-direction signals persisted over the interval in which the fly's turn changed sign; null-direction activity remained comparatively weak.

The results support a transformation downstream of the sampled lobula plate signals. They do not prove that every early neuron lacks relevant dynamics, because the recordings sampled selected populations and calcium activity. Necessity of T4/T5 and localization of the slow behavioral reversal are different experimental conclusions.



Mano et al. (2023), Fig. 5(a–e). T4, T5, HS and CH responses during sustained motion.

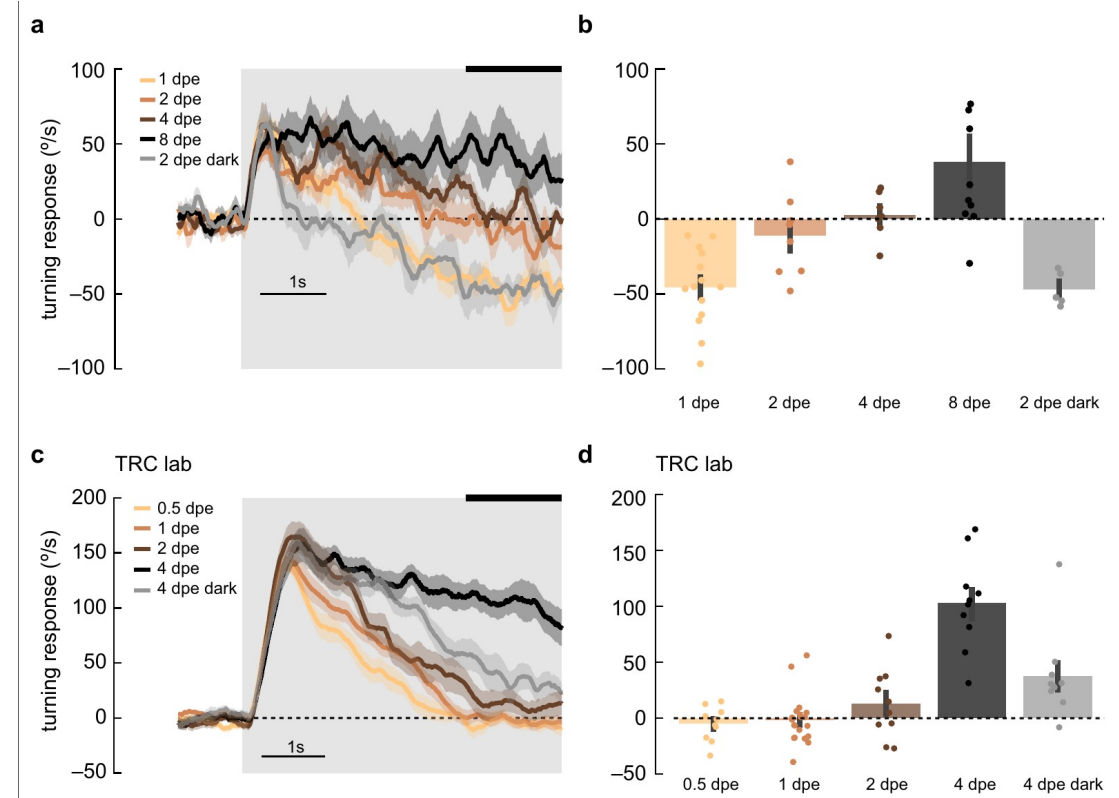
Age and light exposure alter optomotor dynamics

Eclosion is the emergence of an adult insect from its pupal case. Mano and colleagues measured prolonged optomotor turning at different ages after eclosion, testing whether the balance between ordinary and anti-directional responses was fixed in adulthood.

Across 1, 2, 4 and 8 days, *Drosophila melanogaster* shifted toward more sustained turning with the stimulus. Dark-reared flies retained a stronger juvenile-like decrease or reversal.

Comparable experiments in another laboratory also detected age and light-exposure effects.

The finding supports visual-experience-dependent maturation of sensorimotor responses. It does not identify the altered synapse or prove a single learning mechanism. The same article found different age dependence in *Drosophila yakuba*, so plasticity cannot be assumed identical even among closely related fly species.



Mano et al. (2023), Fig. 6. Adult age and dark rearing change prolonged turning.

Key takeaways

Optomotor turning couples panoramic image motion to steering; stimulus contrast, spatial scale and elapsed time affect the response.

The Reichardt detector compares neighboring signals with unequal temporal filtering, nonlinear interaction and opponent subtraction; it specifies a computation rather than a unique two-neuron circuit.

Stimulus structure matters: temporal frequency tuning and reverse-phi responses follow from relationships between space, time and contrast polarity, so local detector output is not a pattern-independent velocity estimate.

Parallel ON and OFF pathways use T4 and T5, respectively; four subtypes of each converge in direction-specific lobula plate layers.

Complementary mechanisms include preferred-direction enhancement and null-direction suppression. Connectomes, transmitter markers, receptor transcripts and voltage/calcium measurements supply distinct kinds of evidence.

Wide-field and motor transformations follow local detection. T4/T5 are necessary for ordinary and reversed turns, while prolonged responses and maturation require additional sensorimotor dynamics.